

Exploratory Data Analysis: Covariation

MKT 566 · Fall 2026

Davide Proserpio

USC Marshall School of Business

Before We Start

Housekeeping

- **Homework 1** is due **Monday, Sept. 14**, on Brightspace. Same format as the case: an R Markdown report knitted to HTML, plus `ai-log.md`
- **Groups**: be in a group by **Tuesday, Sept. 15**. Add yours to the [sign-up sheet](#)
- Today: the solution of the variation case, then **covariation**
- Thursday and next Tuesday: the **RateBeer case** (bring your laptop)

“Am I actually learning anything?”

A common feeling in the first weeks of working this way: the AI writes code you have never seen, you run it, and a chart appears. It feels like magic, not learning. That feeling is **normal**, and it is not a sign that something is wrong.

What this course does not teach

- Writing R code from memory
- Knowing every R function

The AI does that, better than any of us.

What it does teach (and grade)

1. **Choosing** the right chart or analysis for the question
2. **Asking** for it clearly
3. **Reading** the result: does it make sense? what does it say?
4. **Concluding**: what does it mean for the marketing decision?

Running code you could not write yourself is the point, not a problem. The skill is to **read** it and **judge** the result. It takes a few weeks, and the two habits on the next slide speed it up.

Two habits, starting this week

1. Predict before you run

Before you make the chart, write one sentence: *“I expect Search to have the most customers.”*

Then look. Were you right? If not, is it the **data** or the **code**?

Without a guess, every chart looks “fine” and you learn nothing.

2. Explain it back

After the chart works, pick the **one line** of code that does the work. Say in your own words what it does.

Stuck? Ask the AI: *“Explain this line as if I have never coded.”* Then say it back in your words.

If you can explain it, you understand it.

Our next in-class exercise (the RateBeer case, Thursday) has a short box for both under every task. This is the part of the report that shows **you** did the thinking.

Reading ggplot: five pieces, every time

Every chart the AI writes has the same structure:

1	<code>ggplot(df, aes(x = Channel, y = Ad_Spend)) +</code>	# 1. the data, 2. which column on which axis
2	<code>geom_boxplot() +</code>	# 3. the chart type
3	<code>facet_wrap(~ Device) +</code>	# 4. one panel per group (optional)
4	<code>labs(title = "Ad spend by channel",</code>	# 5. labels
5	<code> x = NULL, y = "Ad spend (\$)")</code>	

- `geom_point` = scatter, `geom_col`/`geom_bar` = bars, `geom_histogram`, `geom_boxplot`, `geom_line`, `geom_tile` = heatmap
- Everything after `labs()` is style (`theme_minimal()`, colors, scales)

You do not need to **write** this. You need to **recognize** it. Then, when the AI uses a `geom_` you did not ask for, or puts the wrong column on `x`, you will notice.

The Variation Case: Debrief

Tasks 1–2: what is a row, and what do the summaries say?

```
1 df <- read.csv("data/marketing_eda.csv")
2 summary(df[, c("Age", "Ad_Spend", "Clicks", "Purchases", "Revenue")])
```

Age	Ad_Spend	Clicks	Purchases	Revenue
Min. :18.00	Min. : 1.83	Min. : 0.00	Min. : 0.000	Min. : 0.00
1st Qu.:31.00	1st Qu.: 96.86	1st Qu.: 10.00	1st Qu.: 0.000	1st Qu.: 0.00
Median :43.00	Median : 172.25	Median : 19.00	Median : 1.000	Median : 26.86
Mean :42.51	Mean : 216.78	Mean : 24.17	Mean : 1.047	Mean : 41.21
3rd Qu.:54.00	3rd Qu.: 278.74	3rd Qu.: 32.00	3rd Qu.: 2.000	3rd Qu.: 61.28
Max. :65.00	Max. :3875.35	Max. :322.00	Max. :17.000	Max. :702.61

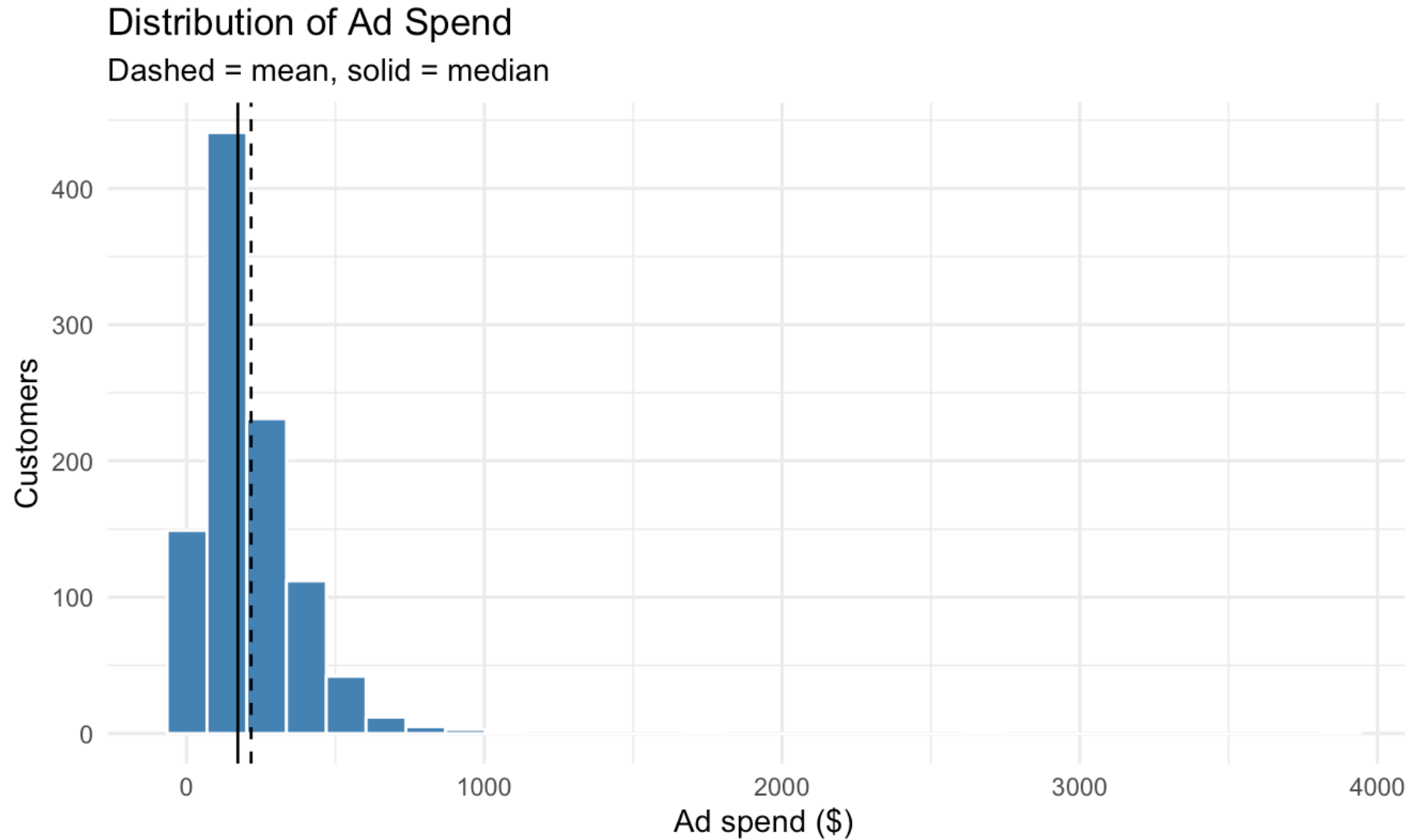
One row = one customer. What did you conclude from this table?

Tasks 1–2: reading it

Age		Ad_Spend		Clicks		Purchases		Revenue	
Min.	:18.00	Min.	: 1.83	Min.	: 0.00	Min.	: 0.000	Min.	: 0.00
1st Qu.:	31.00	1st Qu.:	96.86	1st Qu.:	10.00	1st Qu.:	0.000	1st Qu.:	0.00
Median	:43.00	Median	: 172.25	Median	: 19.00	Median	: 1.000	Median	: 26.86
Mean	:42.51	Mean	: 216.78	Mean	: 24.17	Mean	: 1.047	Mean	: 41.21
3rd Qu.:	54.00	3rd Qu.:	278.74	3rd Qu.:	32.00	3rd Qu.:	2.000	3rd Qu.:	61.28
Max.	:65.00	Max.	:3875.35	Max.	:322.00	Max.	:17.000	Max.	:702.61

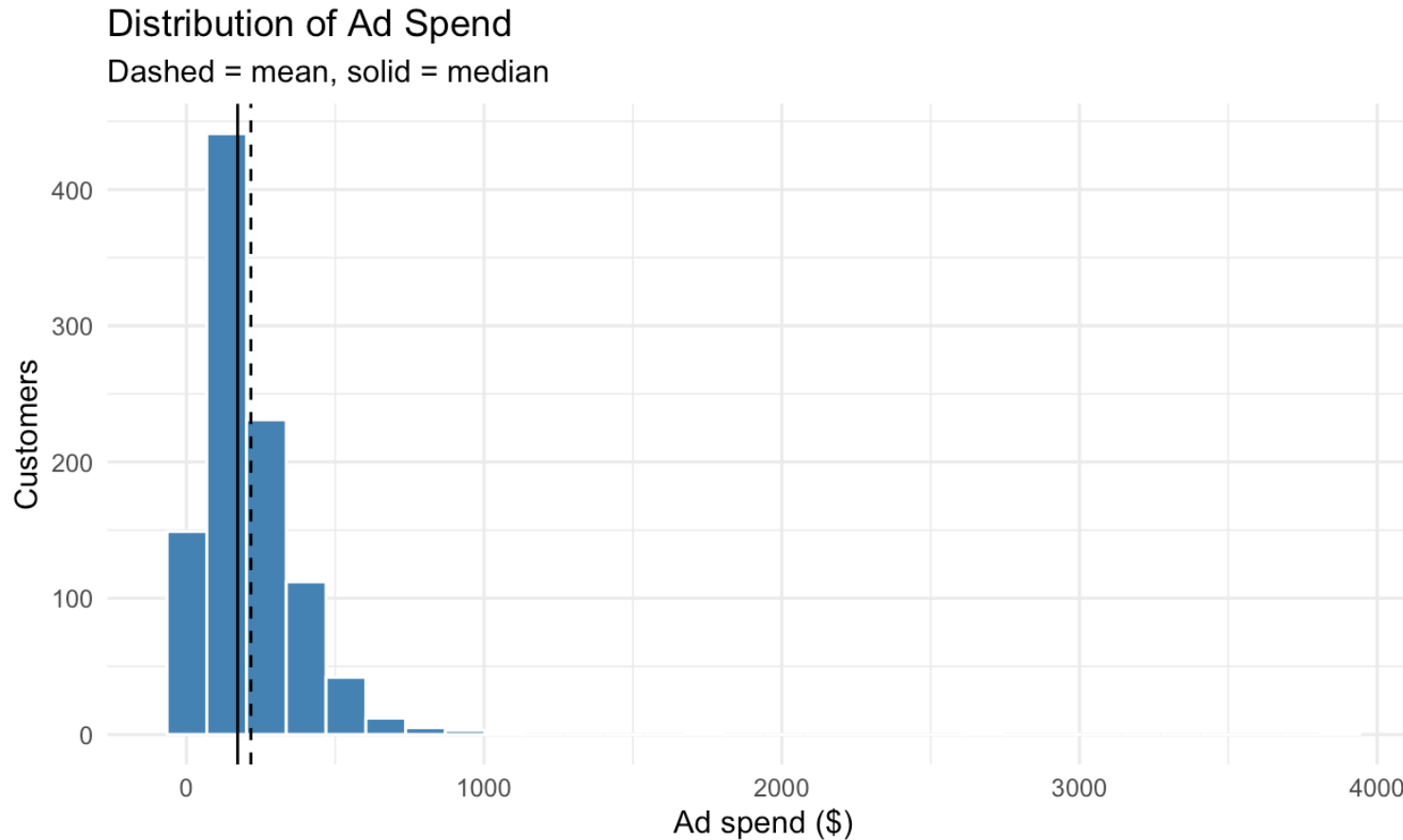
- **No missing values** (there would be an **NA's** row)
- **Ad_Spend**: mean **217**, median **172**. When the mean is above the median, there is a **right tail**: a few customers get much more spend
- **Purchases**: median **1**, max **7**. Most customers buy once or never

Task 3: the distribution of ad spend



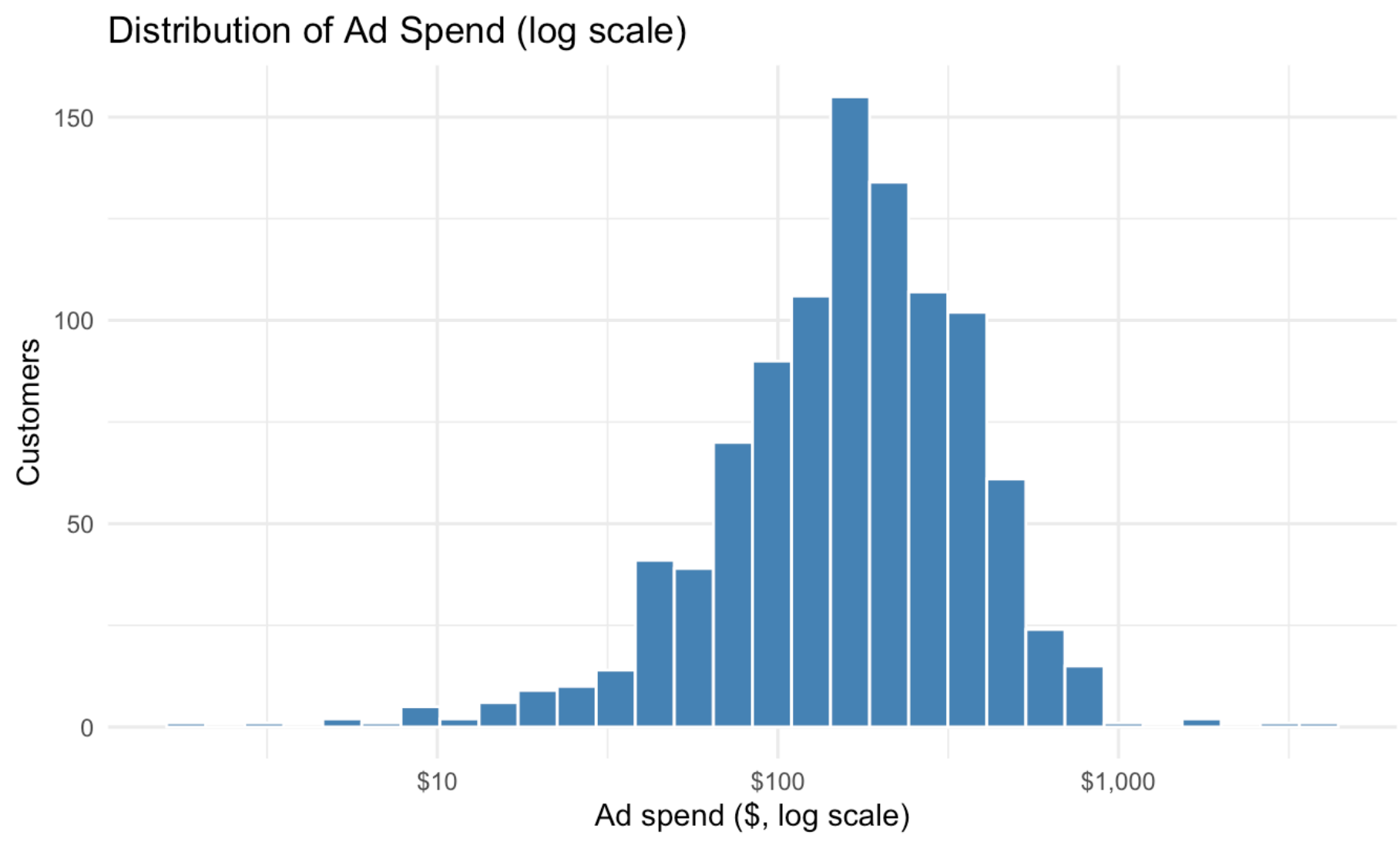
What shape is this, and what marketing strategy could produce it?

Task 3: reading it



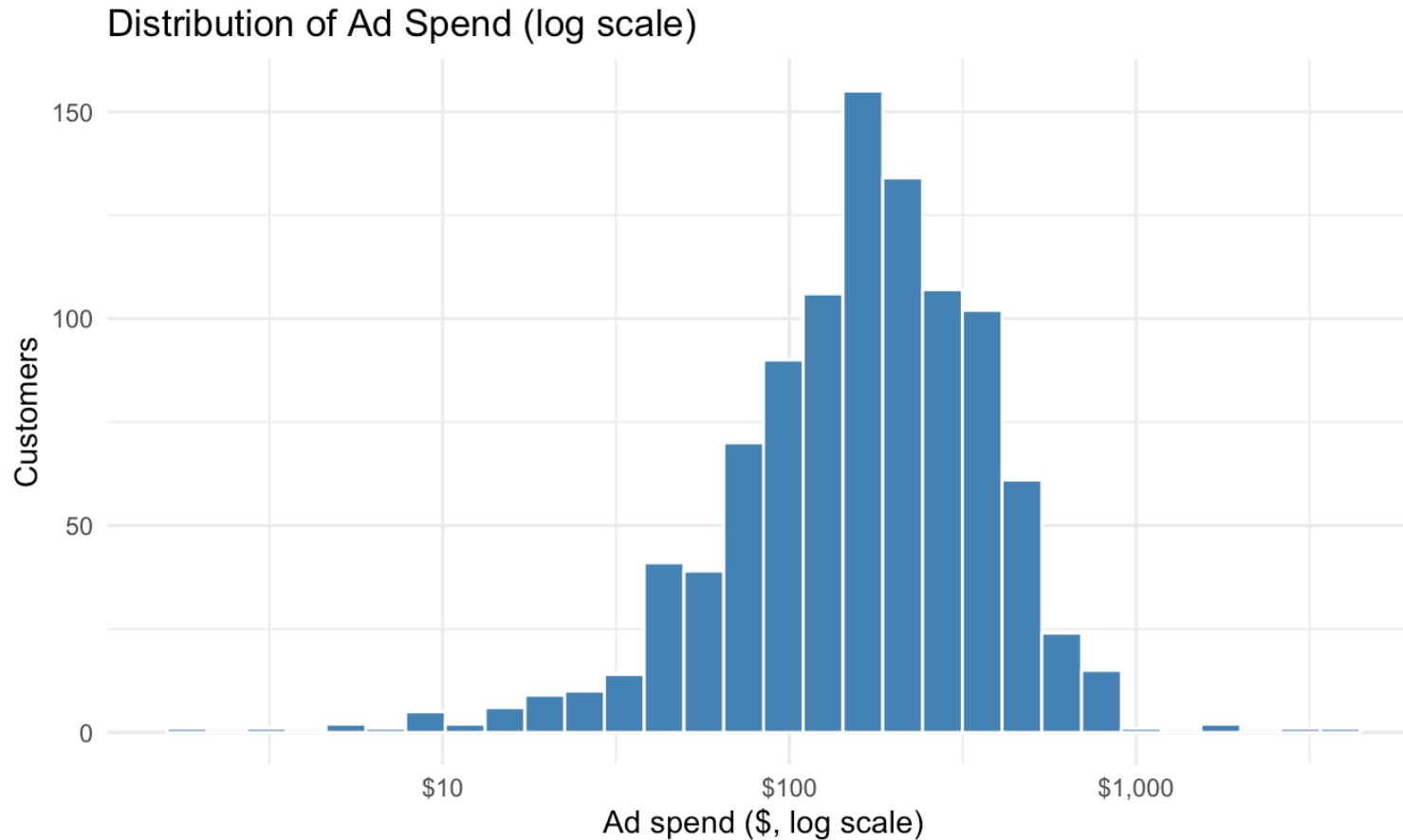
- **Right-skewed**: the mean is to the right of the median, as the summary table suggested
- A likely strategy: a **basic** spend on everyone, plus extra budget on a few customers the firm thinks are high-value (retargeting, lookalike audiences)

Task 4: the same chart on a log scale



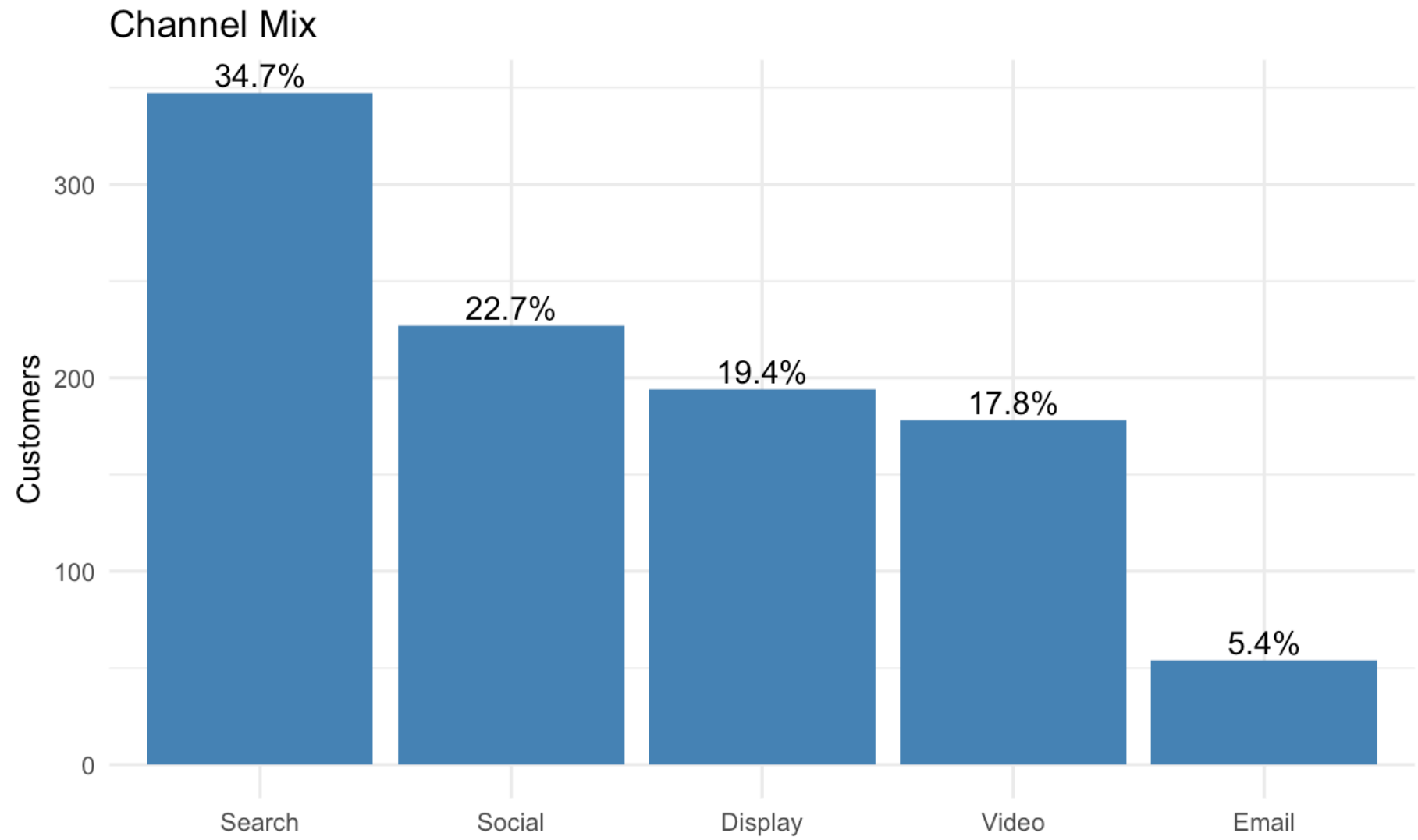
What can you see now that was invisible before?

Task 4: reading it



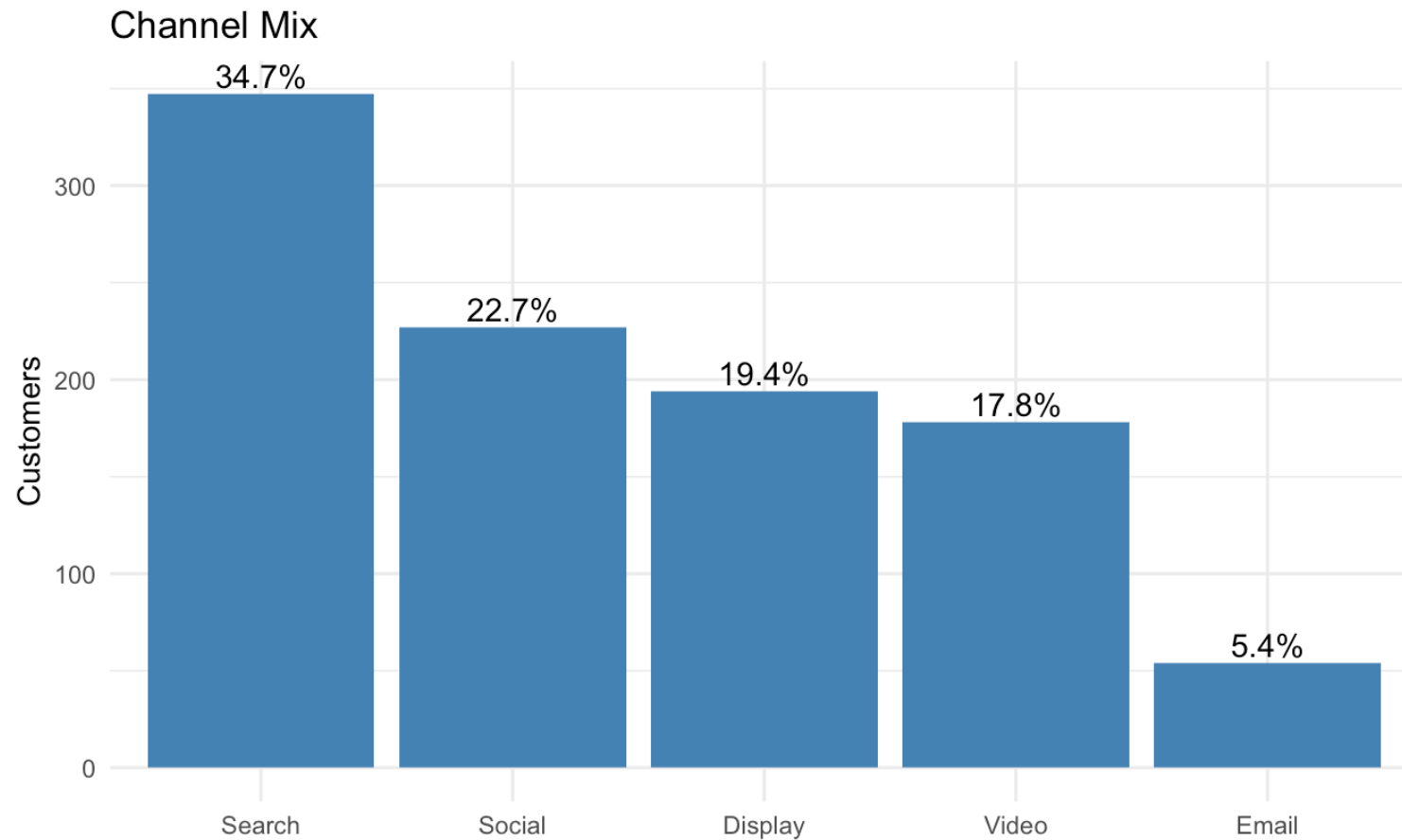
- On the log scale the distribution is roughly **symmetric**. The “typical” customer is easy to see: **\$150–200**
- The low-spend customers were all piled up on the left before. Now they are spread out. Logs stretch small values and squeeze large ones

Task 5: the channel mix



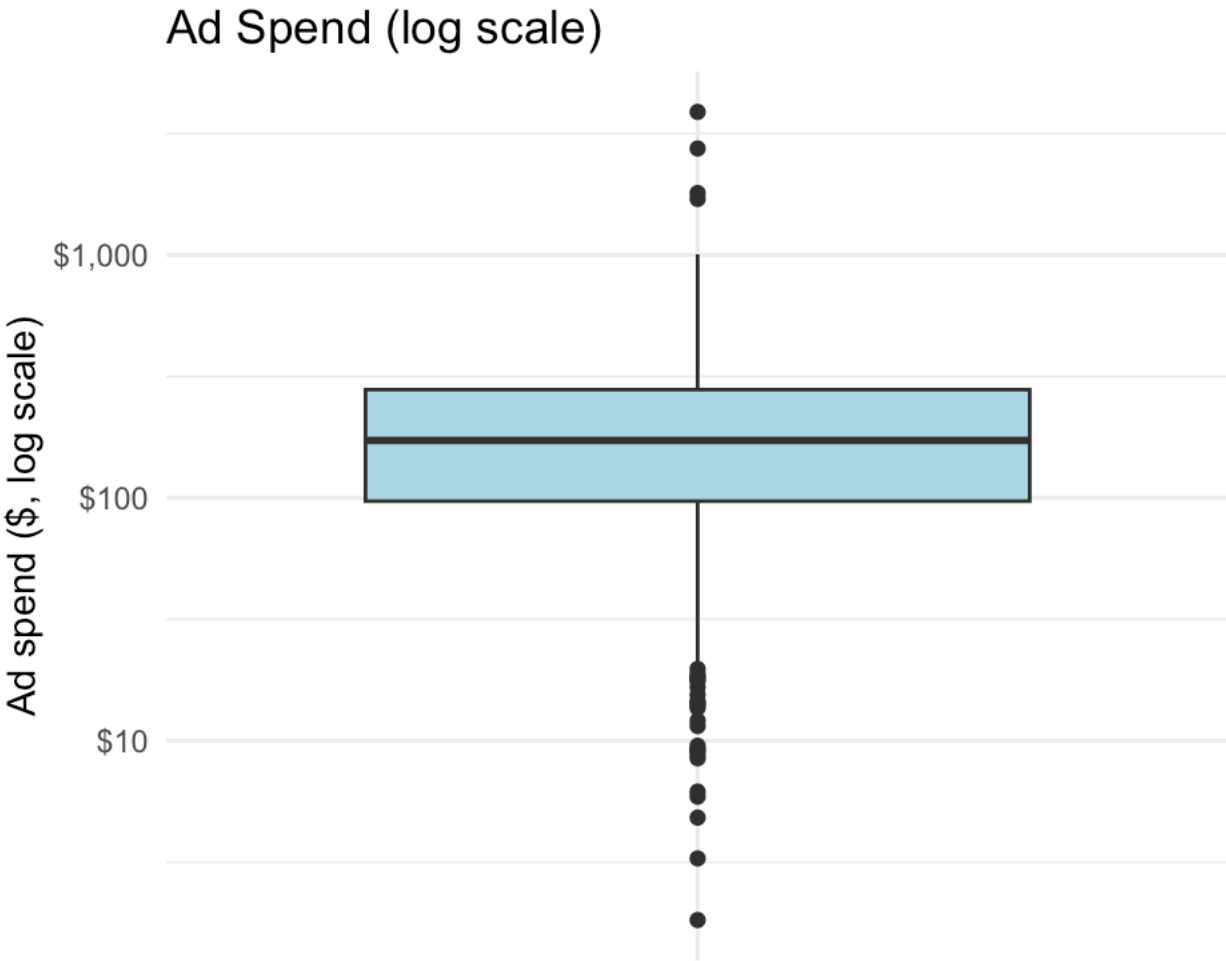
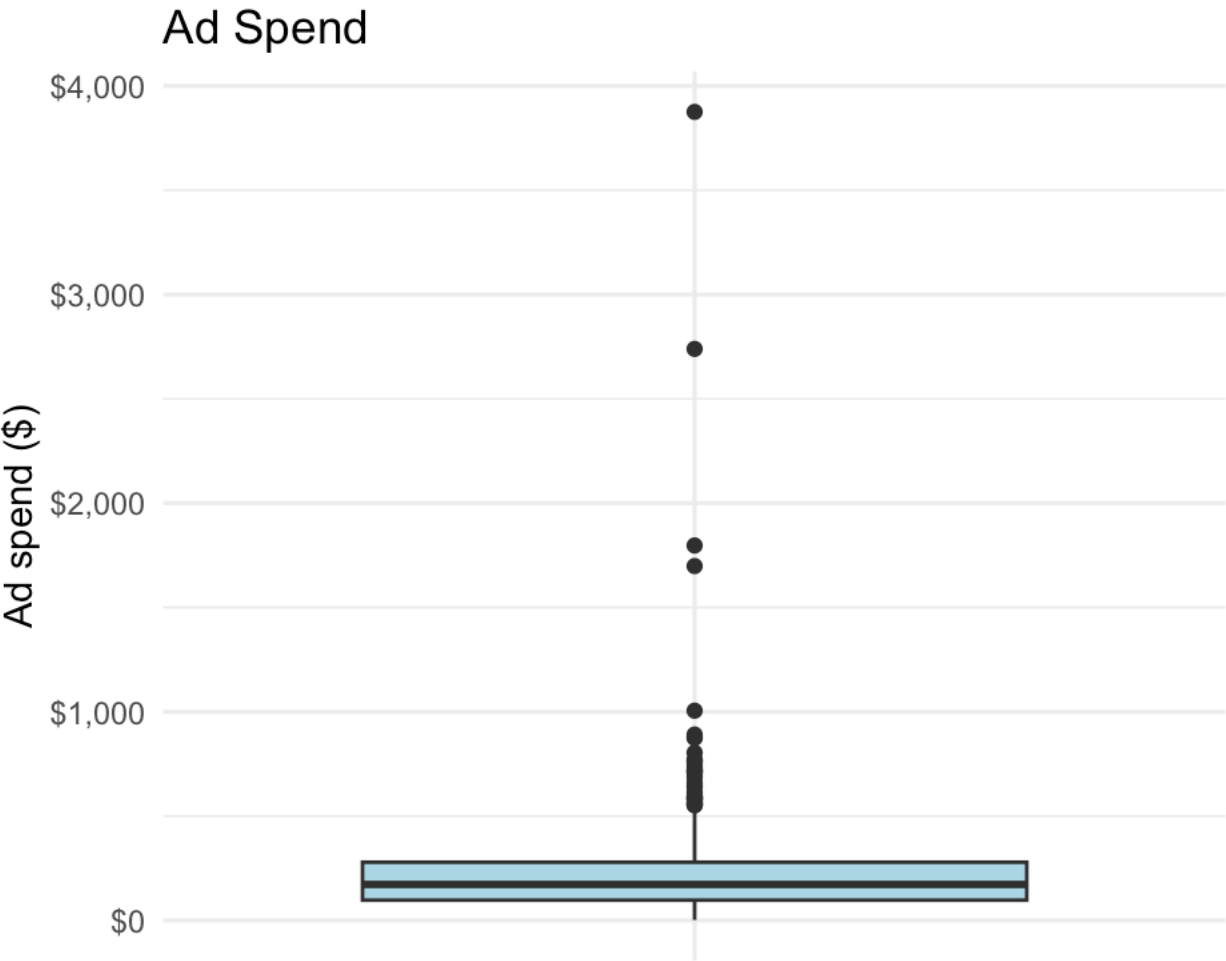
Balanced? Does the mix make sense for an online retailer?

Task 5: reading it



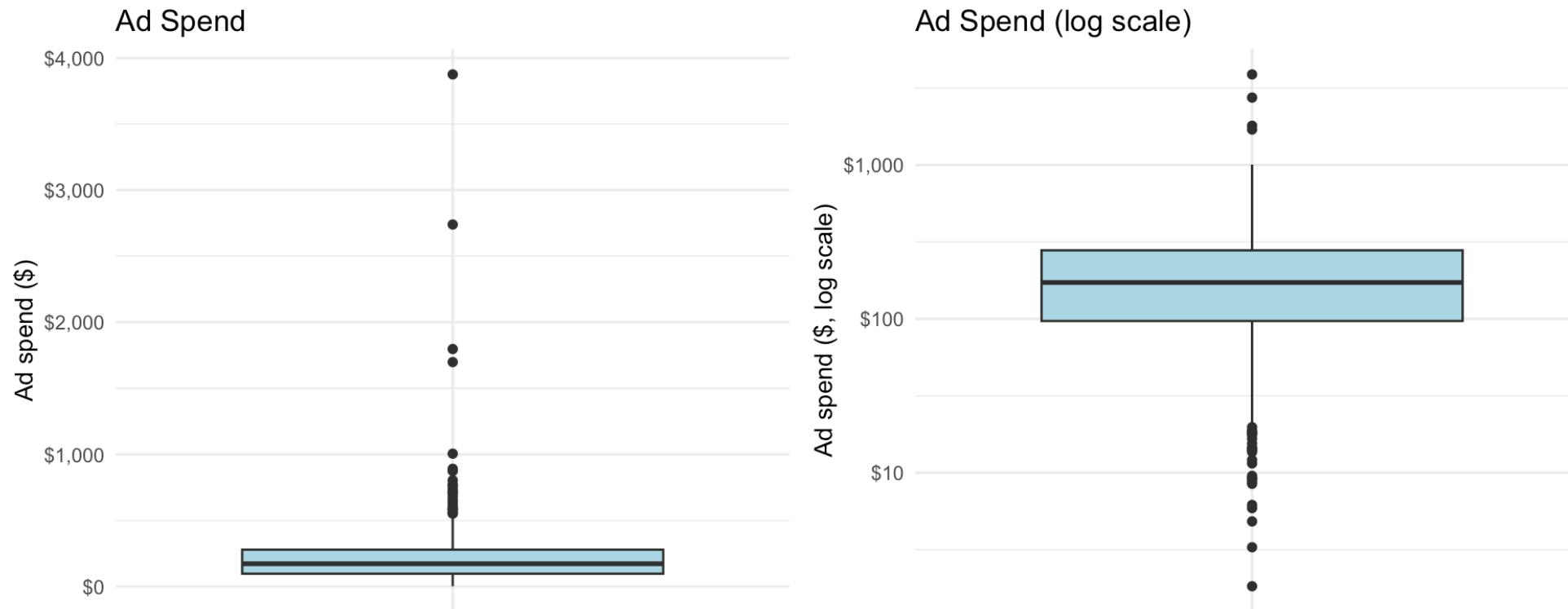
- **Not balanced:** Search is a third of customers, Email one in twenty
- It makes sense: search reaches customers who already **want to buy**, social and display **find new customers**, and email only reaches customers the firm already knows

Task 6: hunting outliers



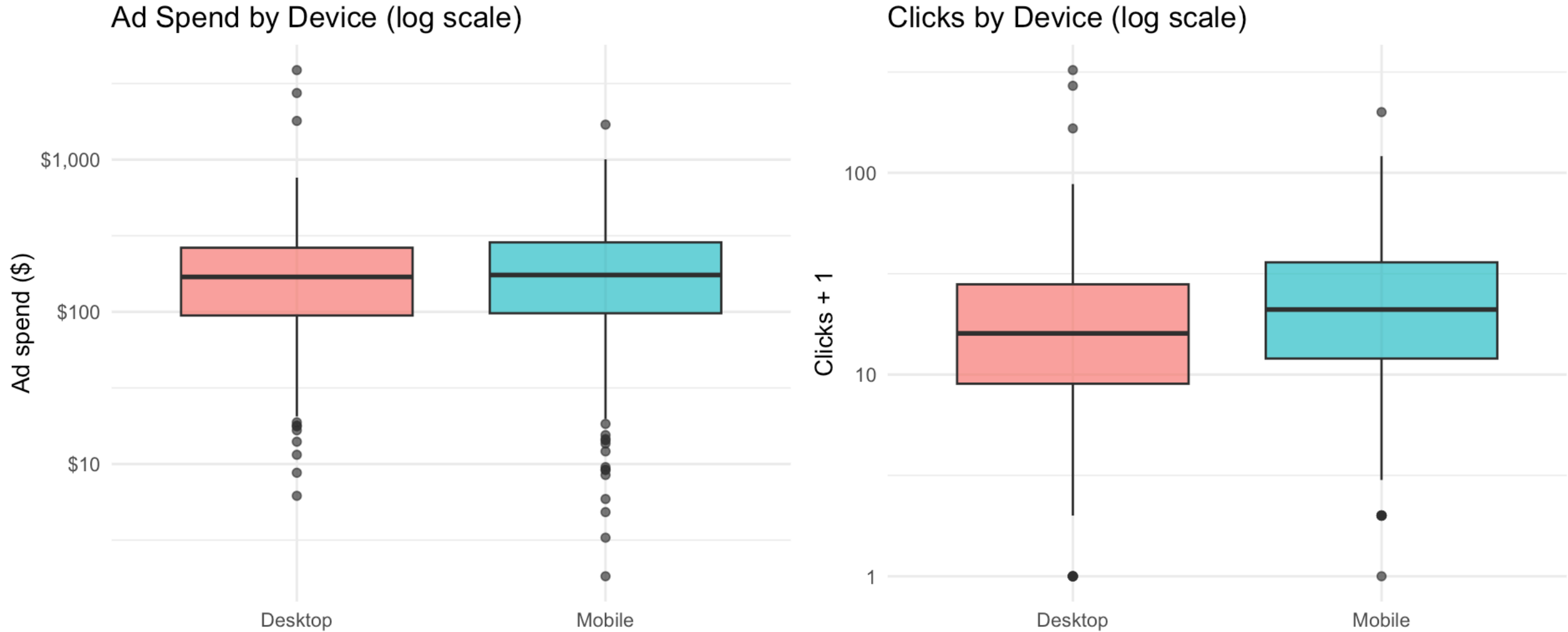
How many outliers, on which side, and should we drop them?

Task 6: reading it



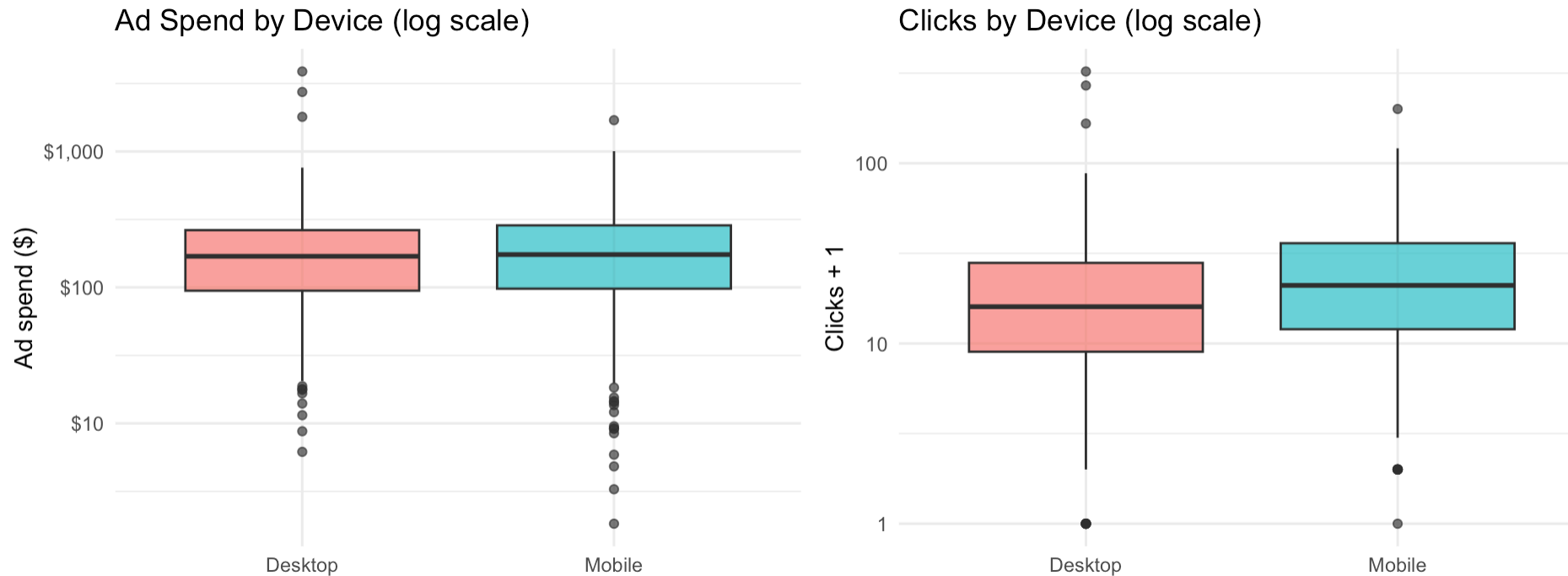
- The $1.5 \times \text{IQR}$ rule flags **39** customers, all on the **high** side. On the log scale only a few remain. Most “outliers” are just the tail of a skewed distribution, not errors
- **Do not drop them** without a reason. Run later analyses **with and without** them, say what you did, and prefer medians to means

Task 7: your own question (two examples)



Same two groups, two different variables, both on a log scale. What is different between the two panels?

Task 7: reading it



- **Spend** is the same on both devices; **clicks** are higher on mobile. The firm spends the same but gets more clicks on mobile
- Why “clicks + 1”? Six customers have **zero** clicks, and the log of zero does not exist. Adding 1 keeps them on the chart

What a strong answer looks like

The question (Task 3): *Is the distribution of ad spend symmetric or skewed? What does the mean vs. the median tell you? What marketing strategy might create this shape?*

Weak

“The distribution of ad spend is right-skewed.
There are some outliers.”

True, but it is just a caption. No numbers, no *why*, no *so what*.

Strong

“Right-skewed: the median customer gets **\$172** but the mean is **\$217**, so a few customers get several times the typical spend. The **39** outliers are all high-spend. That looks like a retargeting budget focused on a few customers, not a data error, so I would keep them and report medians.”

The recipe: **the pattern → the number that shows it → the marketing reason → what you would do**. Four steps, three sentences.

Full instructor solution: [w2-1-variation-case-solution.html](#)

R script for today

In the [code/](#) folder on the course website:

1. [w3-1-eda-covariation-class.R](#): reproduces every chart in the rest of today's deck, with a comment above each one
2. [data/marketing_eda.csv](#): the week 2 case dataset, which we use again today

Download everything as one zip: [w3-code.zip](#), open the folder in VS Code, and run along.

The debrief charts you just saw are in the variation case [solution](#).

Covariation

The two questions of EDA

Last week's framing, one more time:

1. What type of **variation** occurs *within* my variables? (*last week*)
2. What type of **covariation** occurs *between* my variables? (*today*)

- Variation: one variable at a time (its distribution, its trend)
- Covariation: **two or more** variables, and how they change **together**

What is covariation?

Covariation is when two or more variables change together in a related way.

- The best way to see it is to **plot the relationship** between the variables
- Which plot to use depends, again, on the **type** of variables
- Most marketing questions are covariation questions: *does spend increase sales? do mobile users buy more? which channel converts best?*

Which chart for which pair

Pair of variables	Chart
Continuous × continuous (spend, sales)	Scatter plot (+ fitted line), correlation
Categorical × continuous (channel, purchases)	Bars of averages with error bars; boxplots by group
Categorical × categorical (channel, device)	Count table, heatmap
Anything × time	Line chart, one line per group

Use this table when you write prompts: name the pair of variables, and you know what chart to ask for.

Two Continuous Variables

The marketing dataset again

Two hundred campaigns, ad spend on three channels, and sales:

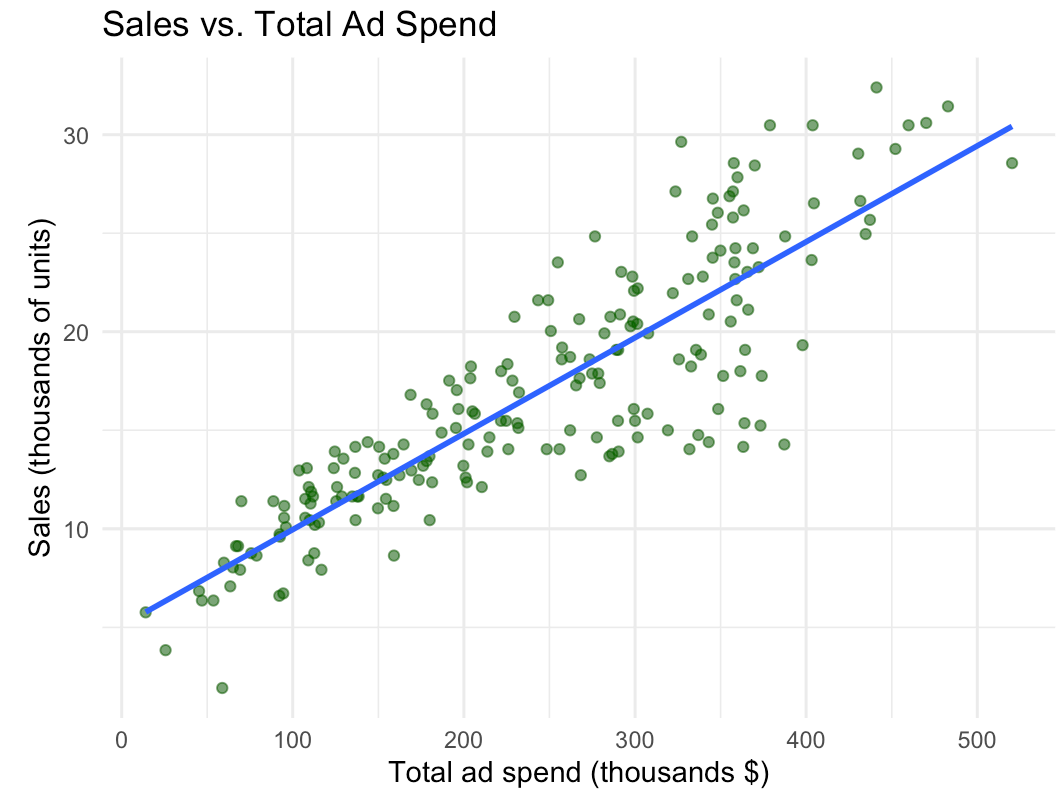
```
1 head(marketing)
```

	youtube <num>	facebook <num>	newspaper <num>	sales <num>	total_ad_spend <num>
1:	276.12	45.36	83.04	26.52	404.52
2:	53.40	47.16	54.12	12.48	154.68
3:	20.64	55.08	83.16	11.16	158.88
4:	181.80	49.56	70.20	22.20	301.56
5:	216.96	12.96	70.08	15.48	300.00
6:	10.44	58.68	90.00	8.64	159.12

How would you visualize sales against ad spend?

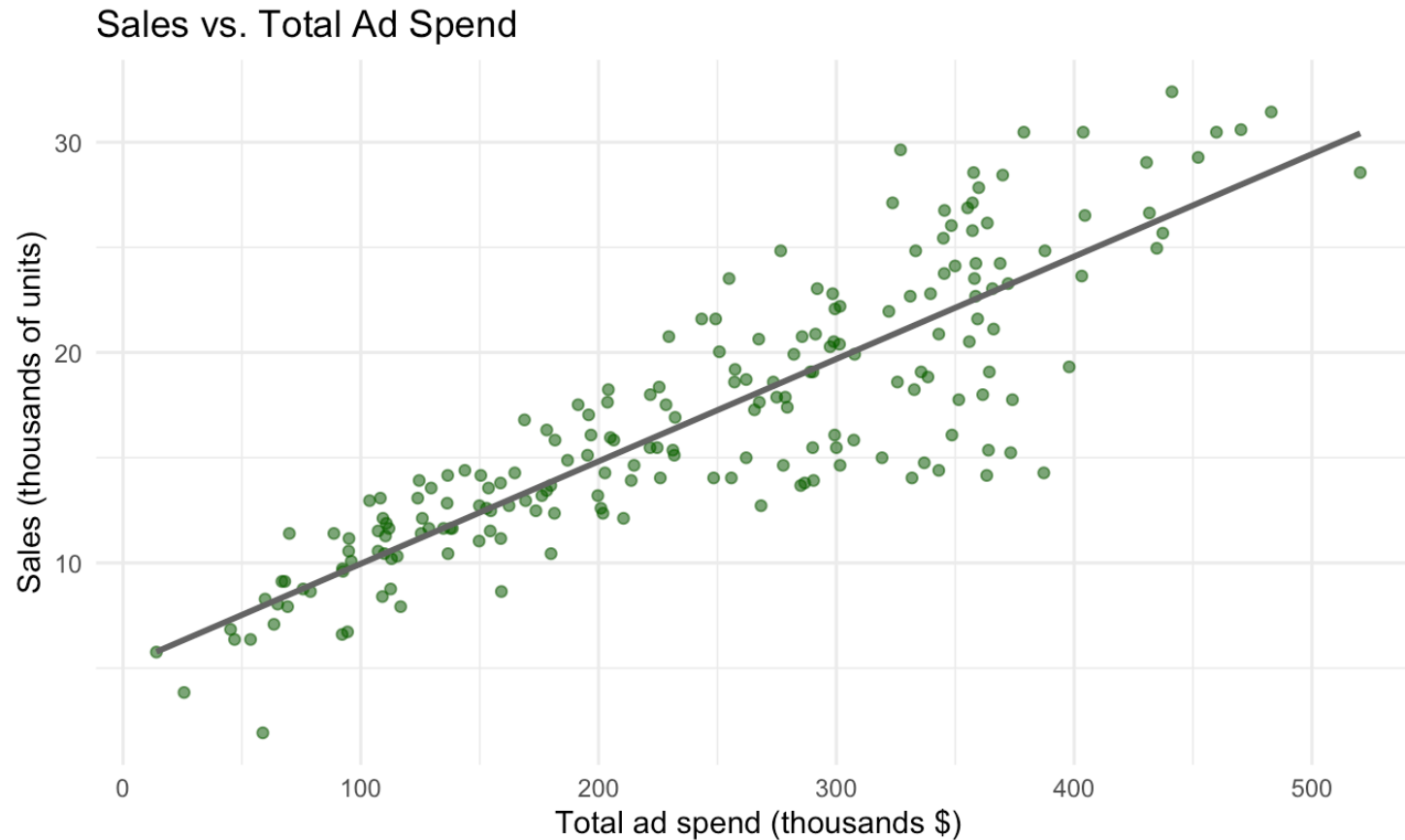
Sales vs. total ad spend

```
1 marketing$total_ad_spend <-  
2   marketing$youtube + marketing$facebook + ma  
3 ggplot(marketing, aes(x = total_ad_spend, y =  
4   geom_point(alpha = 0.6, color = "darkgreen")  
5   geom_smooth(method = "lm", se = FALSE) +  
6   labs(title = "Sales vs. Total Ad Spend",  
7         x = "Total ad spend (thousands $)",  
8         y = "Sales (thousands of units)") +  
9   theme_minimal()
```



Each point is one campaign. What do you see?

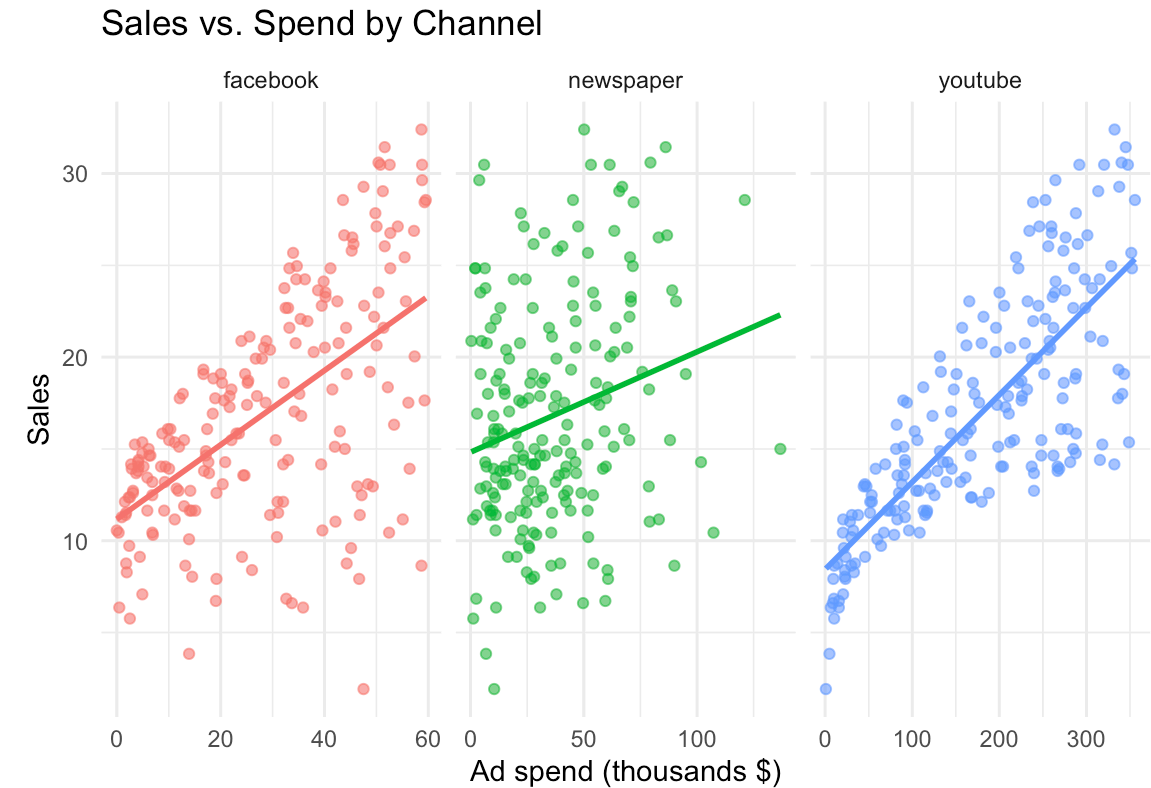
Sales vs. total ad spend: reading it



- A clear **positive** relationship: campaigns that spend more sell more
- Roughly a **straight line**, with a few high-spend campaigns that sold little
- The fitted line (`geom_smooth(method = "lm")`) is a preview of next week: **regression**

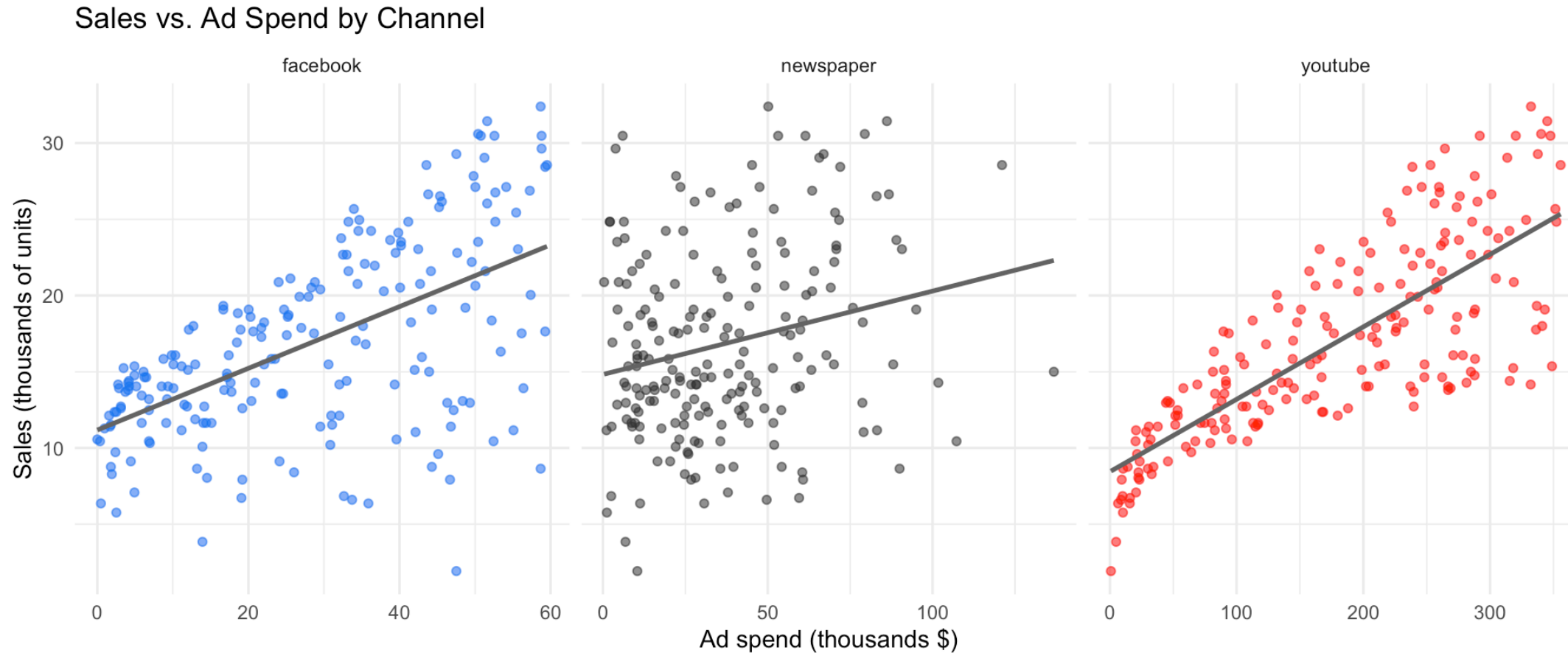
Can we do a more informative viz?

```
1 marketing_long <- pivot_longer(marketing,  
2   cols = c(youtube, facebook, newspaper),  
3   names_to = "channel",  
4   values_to = "ad_spend")  
5 ggplot(marketing_long,  
6   aes(ad_spend, sales, color = channel))  
7   geom_point(alpha = 0.6) +  
8   geom_smooth(method = "lm", se = FALSE) +  
9   facet_wrap(~channel, scales = "free_x") +  
10  labs(title = "Sales vs. Spend by Channel",  
11       x = "Ad spend (thousands $)",  
12       y = "Sales") +  
13  theme_minimal() +  
14  theme(legend.position = "none")
```



Same 200 campaigns, one panel per channel. What changed?

One panel per channel: reading it



- **YouTube:** strong, tight relationship. **Facebook:** positive but noisier. **Newspaper:** almost flat
- The total-spend chart **hid** this difference. Splitting a chart by a categorical variable (`facet_wrap`) is the most useful move in EDA

Correlation: the relationship as one number

The **correlation coefficient** r measures how close two variables are to a straight-line relationship, from -1 (perfect negative) through 0 (none) to $+1$ (perfect positive):

```
1 cor(marketing$total_ad_spend, marketing$sales)
```

```
[1] 0.8677123
```

```
1 cor(marketing$youtube, marketing$sales)
```

```
[1] 0.7822244
```

```
1 cor(marketing$facebook, marketing$sales)
```

```
[1] 0.5762226
```

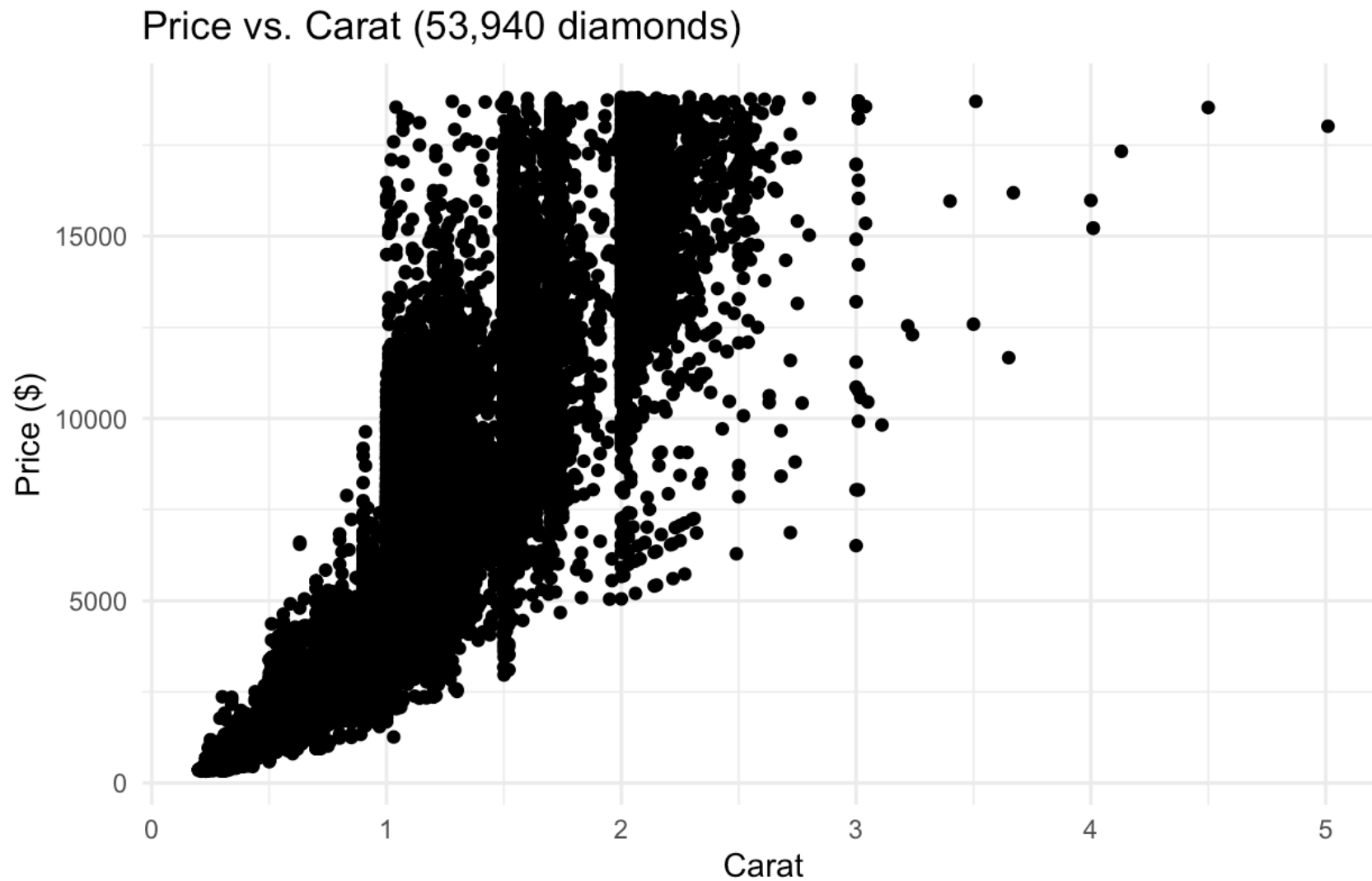
```
1 cor(marketing$newspaper, marketing$sales)
```

```
[1] 0.228299
```

- Same ranking as the panels: YouTube (0.78), Facebook (0.58), Newspaper (0.23)

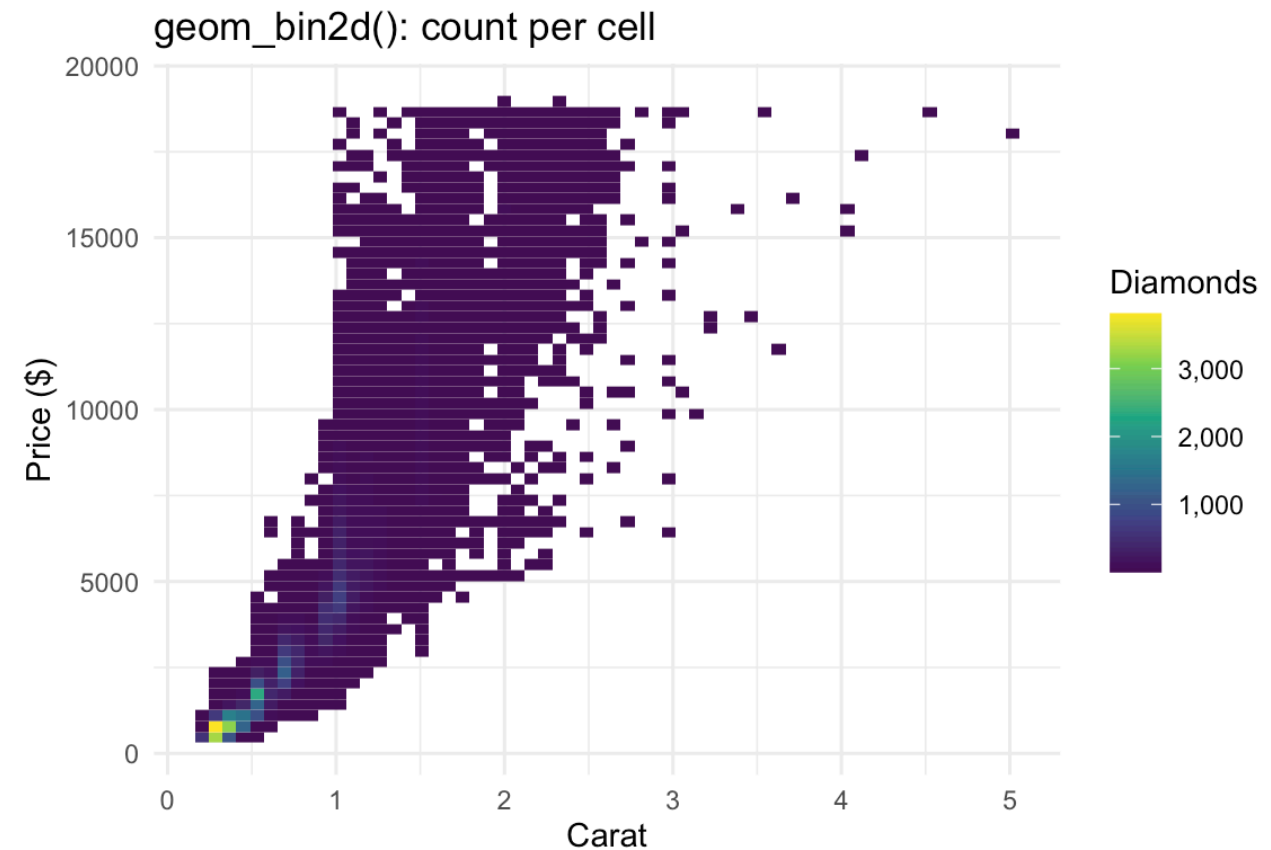
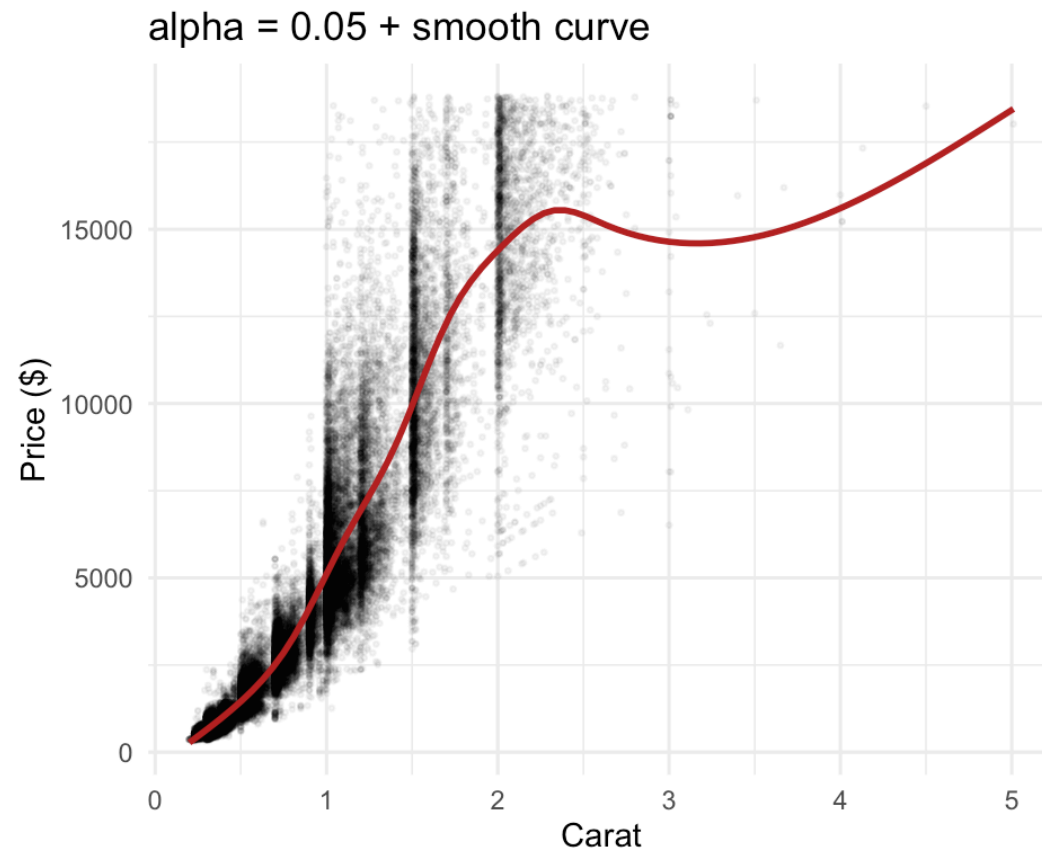
One number hides a lot. Anscombe's quartet is the classic example: four datasets with the same means, the same $r = 0.82$, and the same fitted line, and four completely different pictures. **Plot first, then summarize.**

Overplotting: when you have too many points



ggplot2's [diamonds](#) dataset, 53,940 rows. What is the problem with this chart?

Overplotting: two fixes



- **Transparency** (`alpha`) + a smooth curve shows where most points are and the curved shape of the relationship
- **2D bins** (`geom_bin2d`) count the points in each cell: a heatmap. Use either one when a scatter plot turns into a blob (RateBeer: 290,000 reviews)

One Categorical and One Continuous Variable

Back to the case data

```
1 df <- read.csv("data/marketing_eda.csv")
2 head(df)
```

	CustomerID	Age	Gender	Device	Channel	Ad_Spend	Clicks	Purchases	Revenue
1	1	54	M	Mobile	Social	718.60	95	6	149.16
2	2	18	F	Mobile	Search	233.00	34	1	22.22
3	3	42	F	Mobile	Search	122.51	18	0	0.00
4	4	27	F	Desktop	Social	198.78	19	1	13.22
5	5	53	F	Mobile	Social	145.19	19	4	150.48
6	6	35	M	Desktop	Video	125.74	9	0	0.00

Which chart would you use to explore purchases by gender?

Purchases by gender

```
1 avg_gender <- df %>%
2   group_by(Gender) %>%
3   summarise(avg_purchases = mean(Purchases),
4             se_purchases = sd(Purchases) / sqrt(n()))
5 ggplot(avg_gender, aes(x = Gender, y = avg_purchases)) +
6   geom_col(width = 0.6) +
7   scale_fill_manual(values = c(F = "#E07A5F", M = "#2E5496")) +
8   labs(title = "Average Purchases by Gender",
9        x = NULL, y = "Average purchases") +
10  theme_minimal() + theme(legend.position = "none")
```



First summarize (one row per group), then chart the summary. What do you conclude?

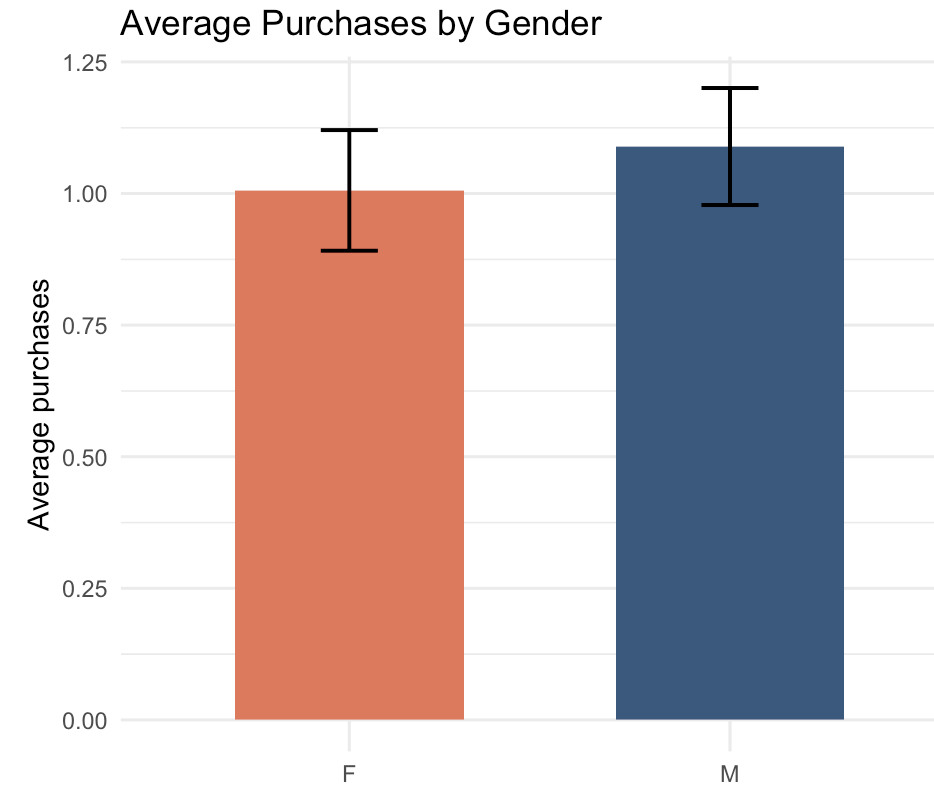
Purchases by gender: reading it



- Men average **1.09** purchases, women **1.01**. Is the difference **real**, or would another 1,000 customers flip it?
- The bars cannot tell you. We need to know how **precise** each average is

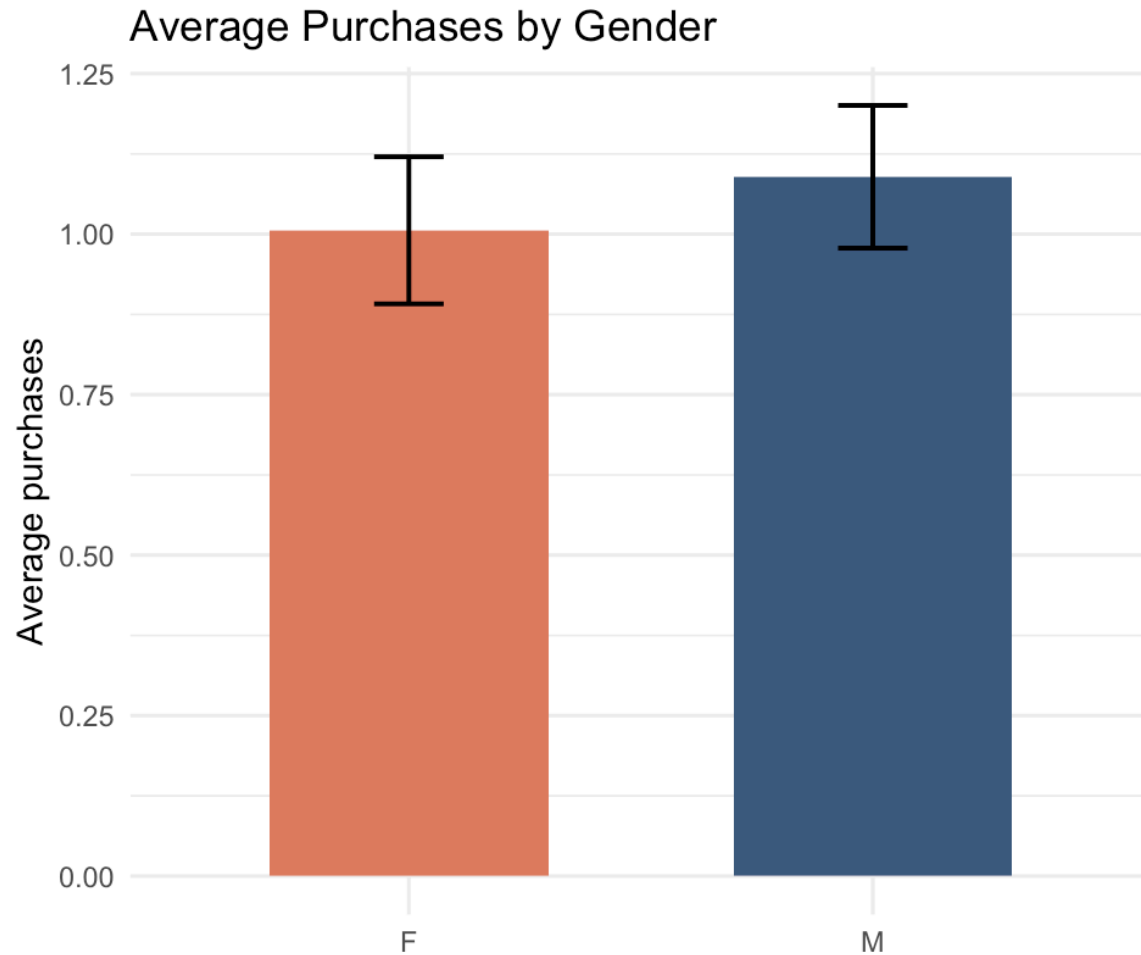
Add error bars

```
1 p_bar +  
2   geom_errorbar(  
3     aes(ymin = avg_purchases - 1.96 * se_purch,  
4         ymax = avg_purchases + 1.96 * se_purch,  
5         width = 0.15, linewidth = 0.7)
```



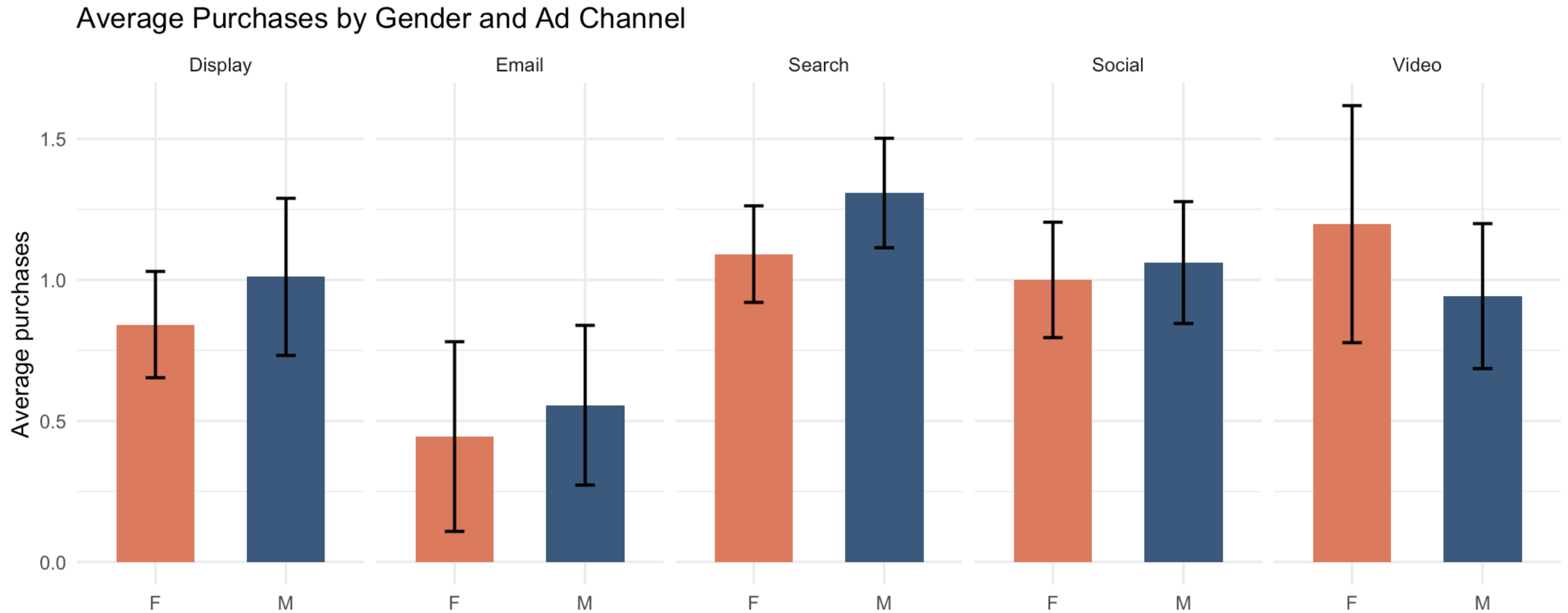
The **standard error**, $SE = sd/\sqrt{n}$, tells us how precise the average is. $\text{Mean} \pm 1.96 \times SE$ is a **95% confidence interval**: the range where the true average most likely is. Now what do you conclude?

Error bars: reading it



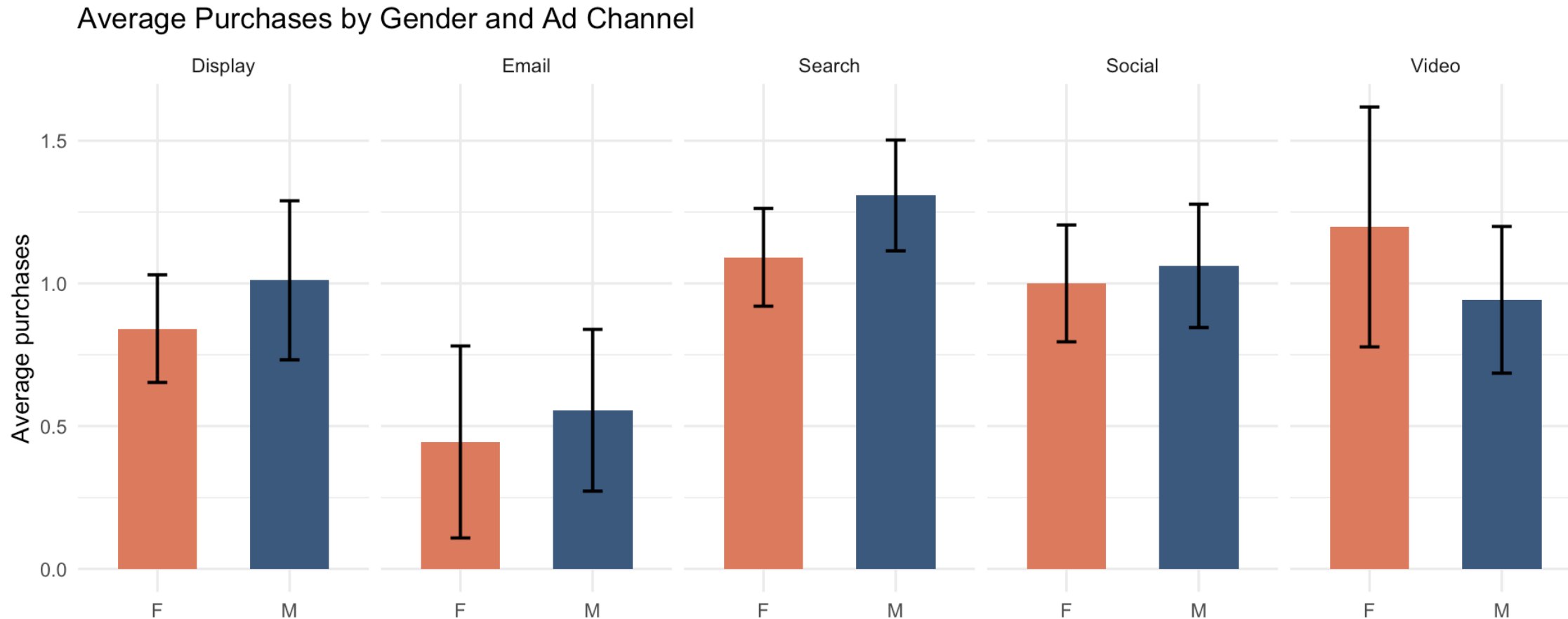
- The two intervals **overlap** a lot: the data cannot tell men and women apart on purchases
- Rule of thumb: bars of averages **without** error bars make small differences look real. Always ask for error bars

Add a third variable: the ad channel



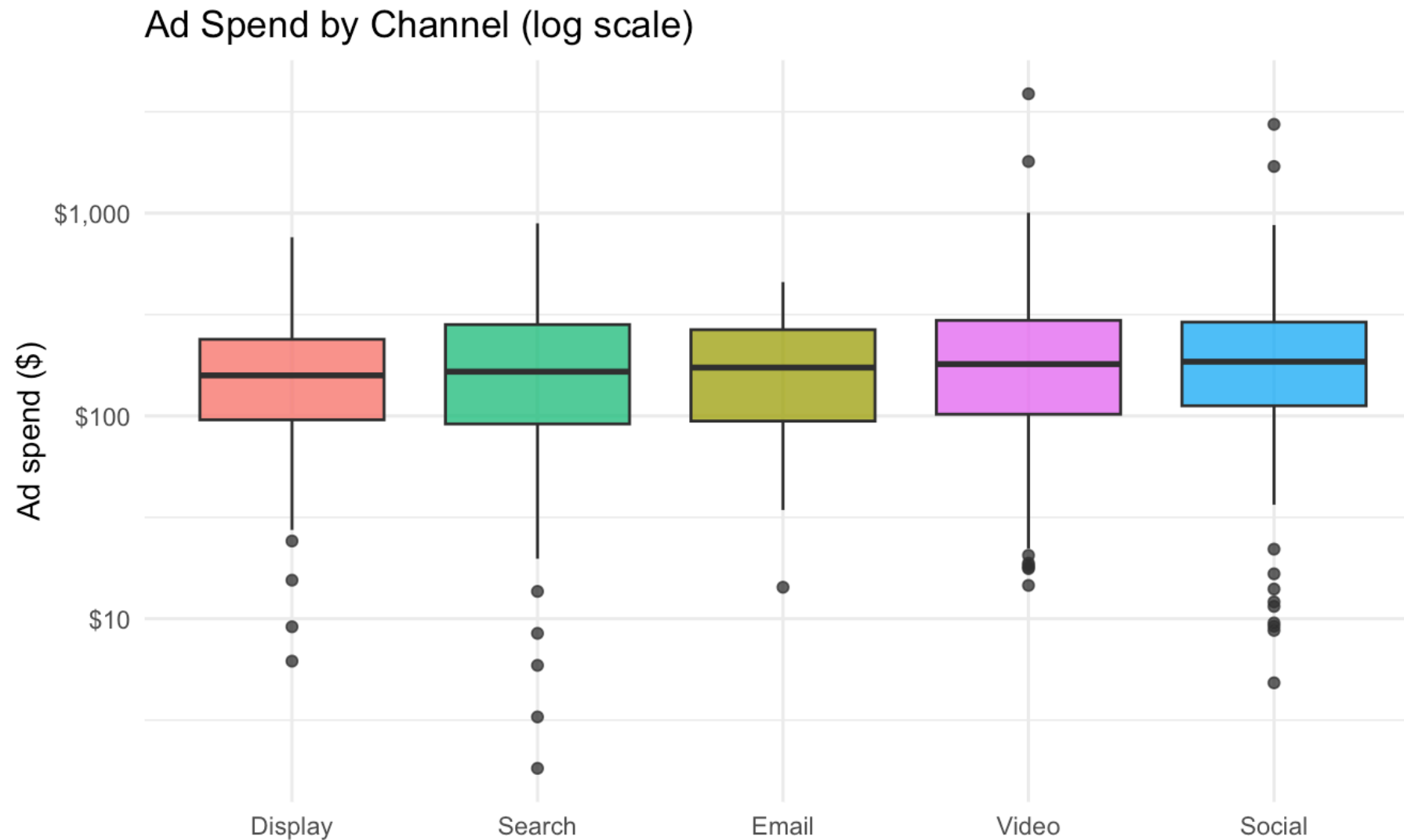
`facet_wrap(~Channel)` again. What do you see now?

Gender × channel: reading it



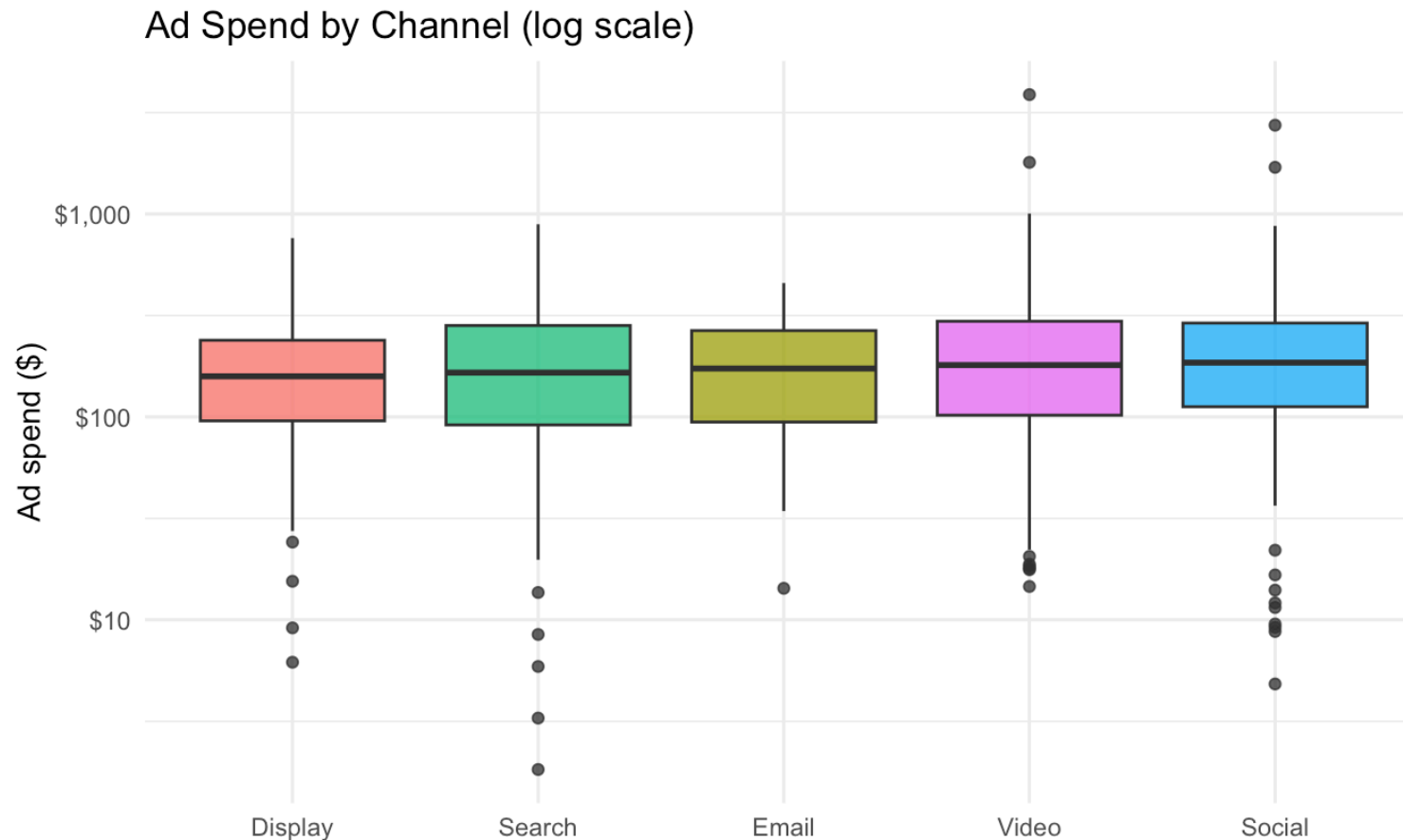
- The gap changes by channel: men buy more via **Search** (1.31 vs. 1.09), women via **Video** (1.20 vs. 0.94)
- **Email** has the widest error bars: only 54 customers. Small group, wide interval. Always check how many people are behind a bar

Averages hide distributions



One boxplot per group, sorted by median. Do the channels differ, and how?

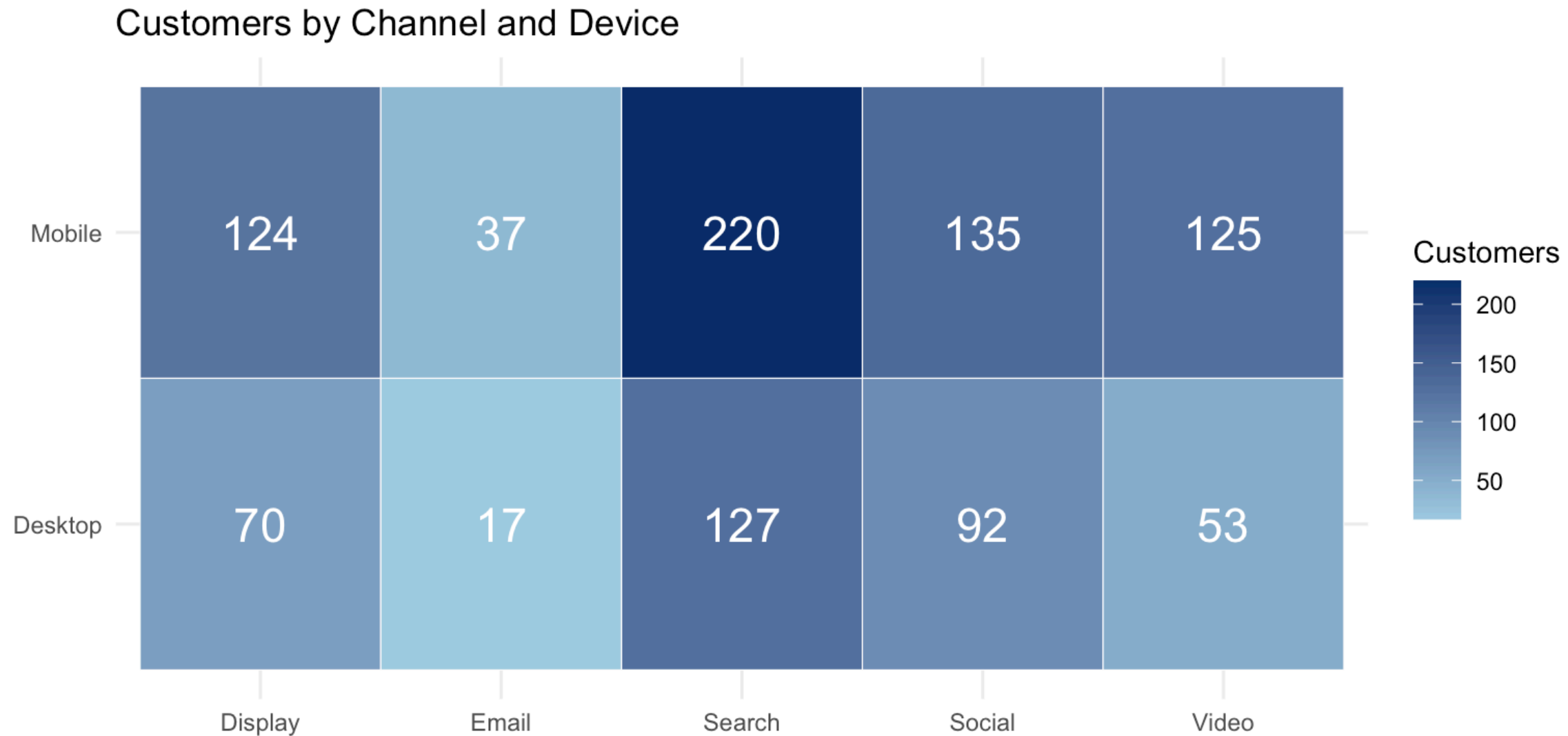
Boxplots by group: reading it



- The **medians** are almost the same: the typical customer gets the same spend in every channel
- The differences are in the **tails**: Video and Social have more high-spend outliers. An average would say “Video spends more”. The boxplot says “Video spends more **on a few customers**”. A different conclusion

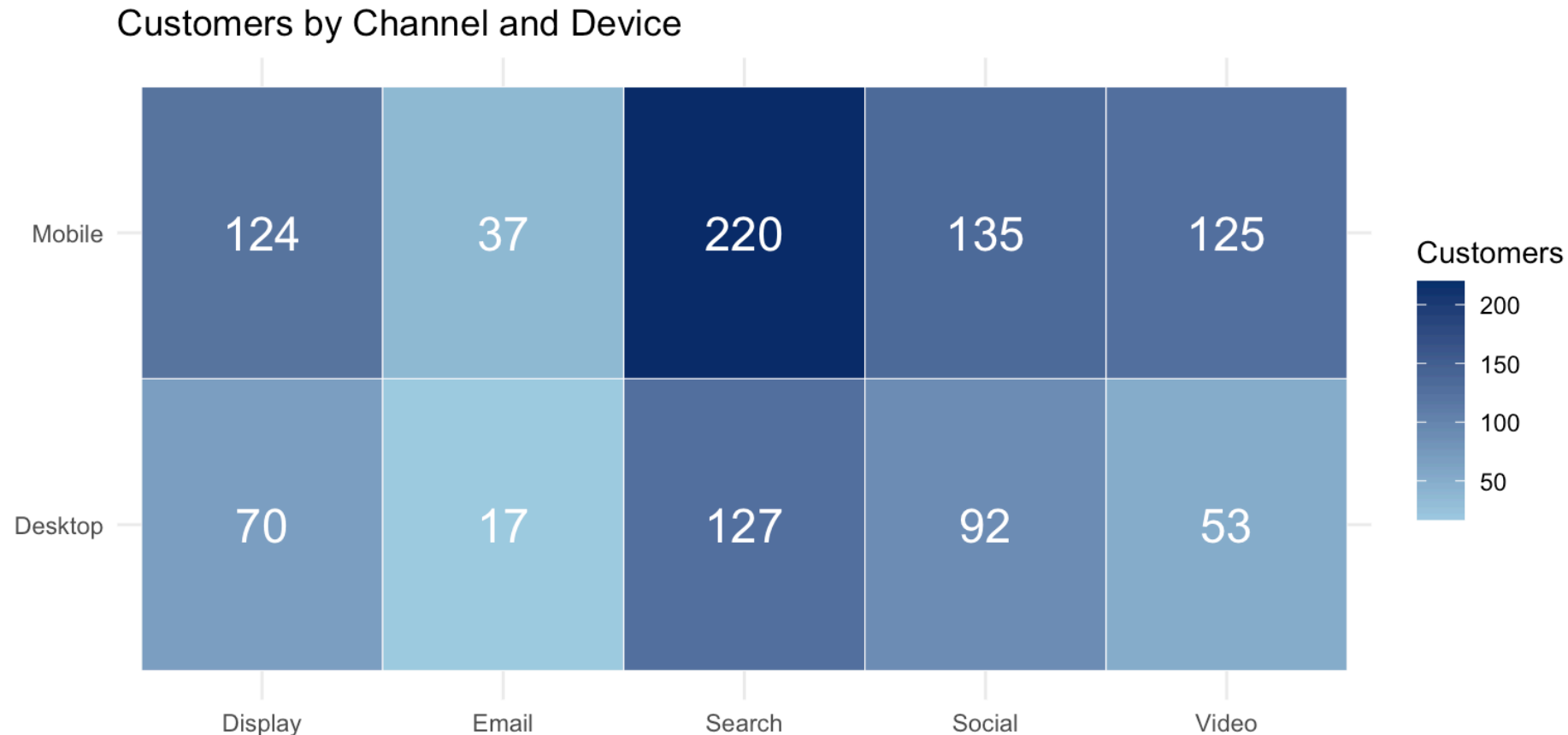
Two Categorical Variables

Channel and device: how many customers in each cell?



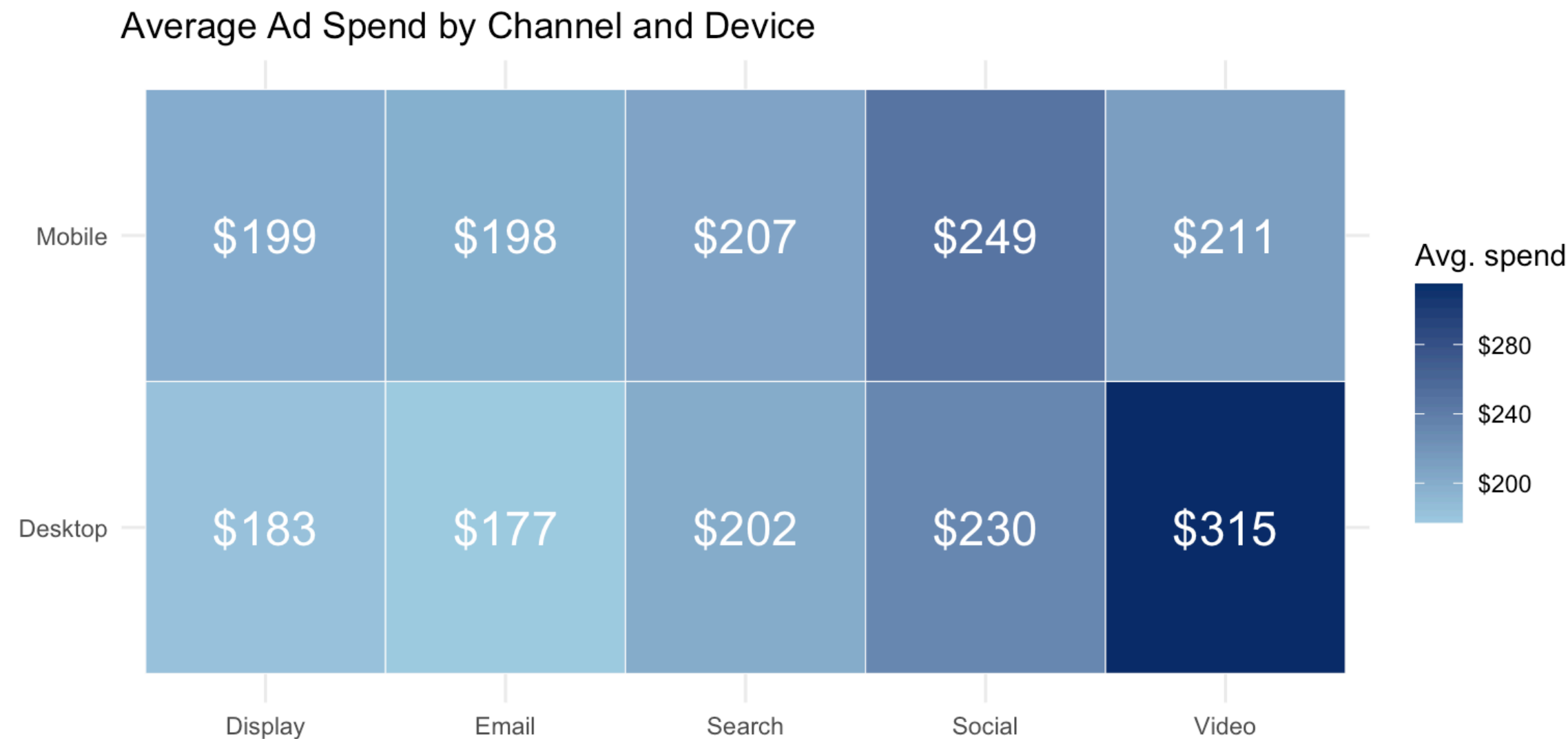
A **heatmap**: color = count, number on top. What do you see?

Counts heatmap: reading it



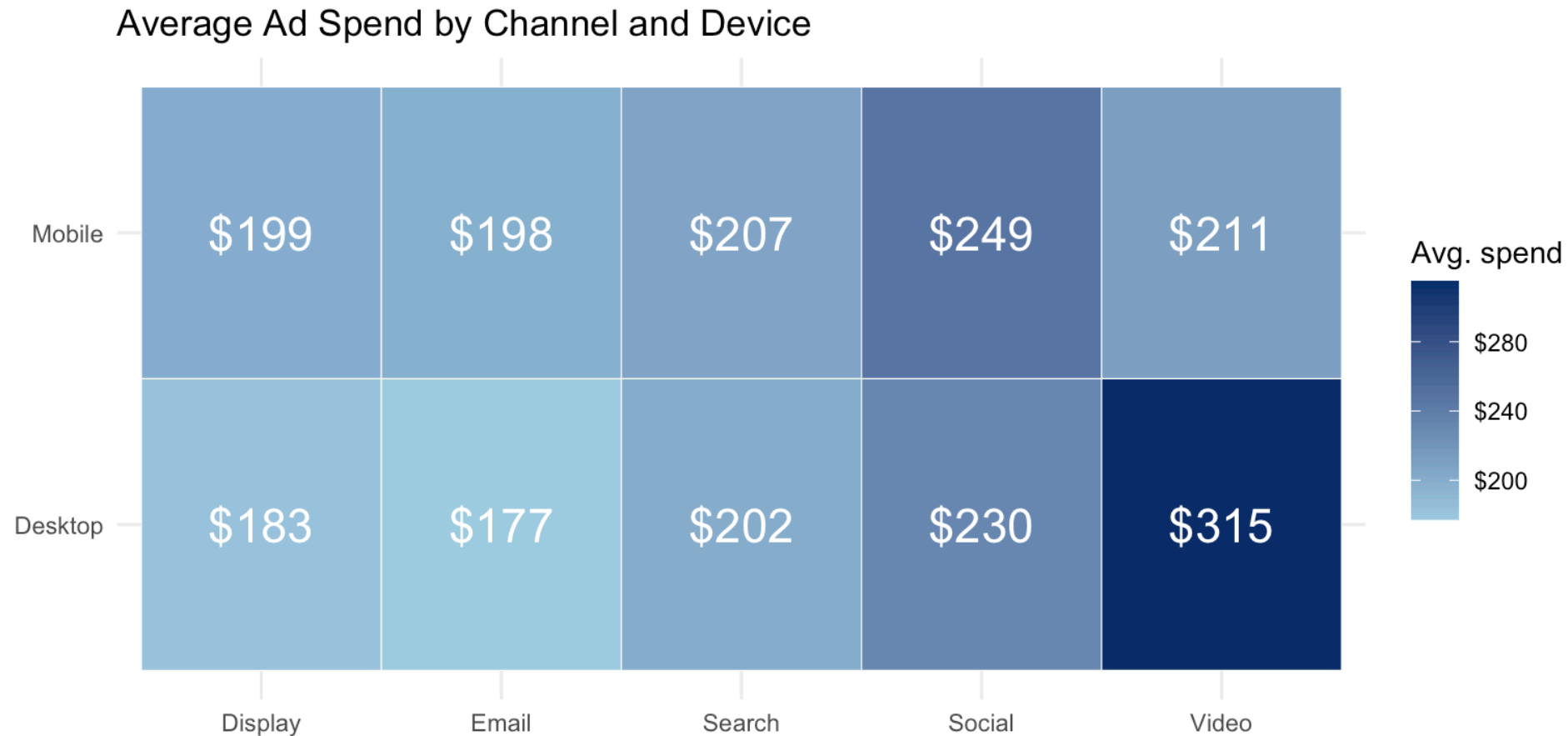
- **Mobile** is the majority in every channel (65% overall)
- **Video** is the most mobile-heavy (125 vs. 53): video ads show up where people watch video
- A heatmap is just a **table with color**. For two categorical variables, that is usually all you need

Add a third variable: average ad spend per cell



Same grid, the color is now the average spend in the cell. Anything stand out?

Spend heatmap: reading it, and the EDA loop



- **Video / Desktop** at \$315 is far above every other cell. Is it a real pattern, or a few outliers in a cell with only **53** customers?
- To find out, go back and make a boxplot of that cell (it is in the class script). This is the EDA loop: **notice** → **ask** → **check** → **refine**

Correlation Is Not Causation

Does ad spend *cause* sales?

Sales and ad spend move together strongly ($r = 0.87$). Three stories fit that fact:

1. **Spend causes sales**: the ads work
2. **Sales cause spend**: managers set budgets as a share of expected sales, so big markets get big budgets *and* big sales
3. **Something else causes both**: the holiday season raises spend and sales at the same time

EDA can show you that two variables **move together**. It cannot tell you **why**. Weeks 11–12 are about telling these stories apart.

Until then, choose your words carefully: “spend **is associated with** sales”, not “spend **drives** sales”.

The RateBeer Case

The case

Thursday you get **292,680 beer reviews** from RateBeer (a beer-review site, now closed), 2000–2012: beer, brewery, style, alcohol content, five ratings, and a date.

The client: a **new craft brewery** deciding **which styles to brew and how many**.

Two things make this harder than last week:

- The data is **dirty**: ratings stored as text like "**14/20**", alcohol content with "**–**" for missing, dates stored as numbers, ciders and sake mixed in with the beers. Finding the problems is your job; fixing them is the AI's
- The questions are about **covariation**: rating by style, rating vs. alcohol, number of styles vs. rating, and how all of this changes over time

How the case works

The handout ([w3-ratebeer-case.html](#)) works like the variation case, with less help:

1. **Tasks 1–3** (look, find the problems, clean): an exact prompt, then prompts with blanks
2. **Tasks 4–8** (variation warm-up, four covariation questions, time): only the goal, you write the prompt
3. **Task 9**: three recommendations to the brewery, each backed by a number

Under every task: **predict** before you run, **explain back** one line after.

Two sessions: **Thursday** Tasks 1–6, **Tuesday, Sept. 15** Tasks 7–9 and the debrief. Deliverable, same as last week: an **R Markdown report knitted to HTML** plus [ai-log.md](#), due **Sunday, Sept. 20**.

Bring your laptop with the week 1 setup working. Download the data ([ratebeer-case-data.zip](#), 5 MB) before class.

Wrap-up

Code and data to reproduce today's charts, on the course website:

- [w3-code.zip](#): one-click download of everything below
- [w3-1-eda-covariation-class.R](#): every chart shown today
- [data/marketing_eda.csv](#): the case dataset from week 2
- [The RateBeer case handout](#) and its [data](#)

Required reading: [Chapter 7 of R for Data Science](#) (the covariation section). Optional: Chapters 3–5 of *R for Marketing Research and Analytics*.

Questions?